Lecture-01

Topics

- About CSE-305
- Why Assembly and 8086
- What is 8086
- System Architecture 8086
  - Registers
  - Segments
- Basic of Assembly Language
  - Program Structure
  - Input Output
  - Basic Instruction
- Emulator

Book: Assembly Language Programming and Organization of the IBM PC

CSE-305 Course Overview

Classroom access details omitted from the public reference copy.

Original PDF page numbering is retained.

User vs Programmer vs Computer Scientist



Using a **developed** software (set of instruction).



Building a software for a **particular architecture** or in a **particular language**.



Building and Design the **system architecture** or the **compiler** or the **interpreter**.

Only by studying assembly language it is possible to gain a feeling for the way the computer "thinks and why certain things happen the way they do inside the computer. High-level languages tend to obscure the details of the compiled machine language program that the computer actually executes."

Why Assembly & 8086

- To understand the **system architecture**, we will see 80X86 as examples
- **To understand how the Higher Level Instruction is Executed by the Hardware**

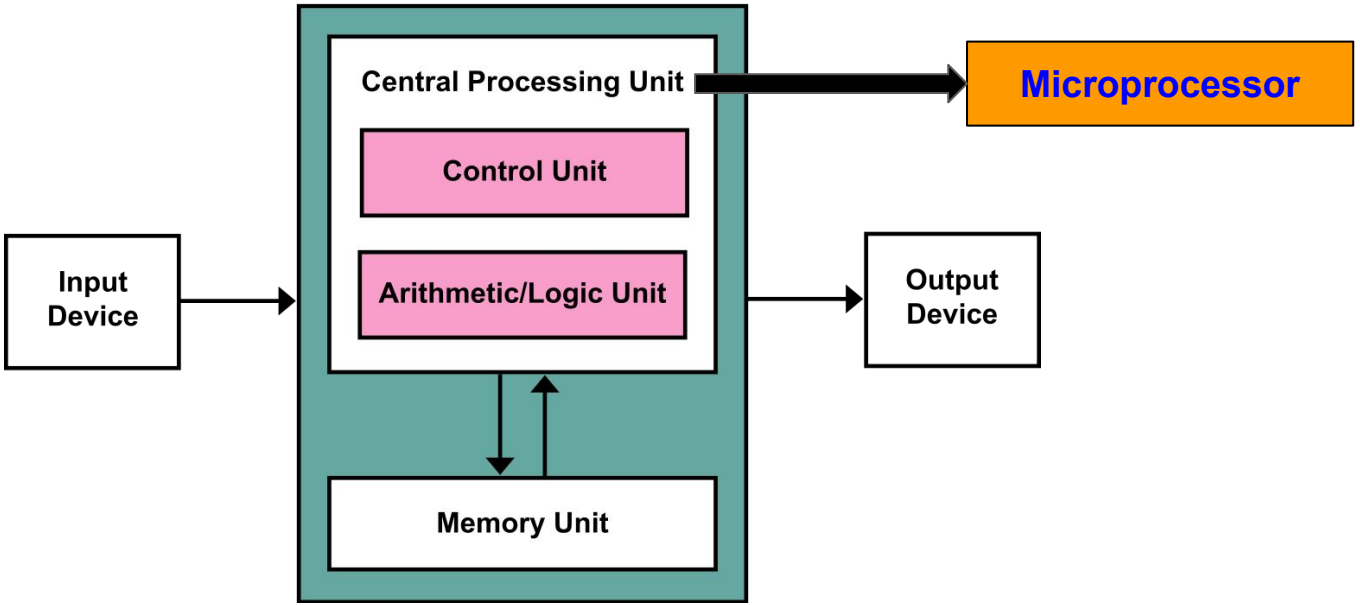
High Level Instruction	Low Level Language	Machine Code
a = a + b	MOV AX, B ADD A, AX	01000111 11010001

- It will help you to design **Compiler** and to understand few concepts of **OS** such as **Atomic Instruction, Mutex etc.**
- Finally, To understand the insight of the computer



What is 8086?

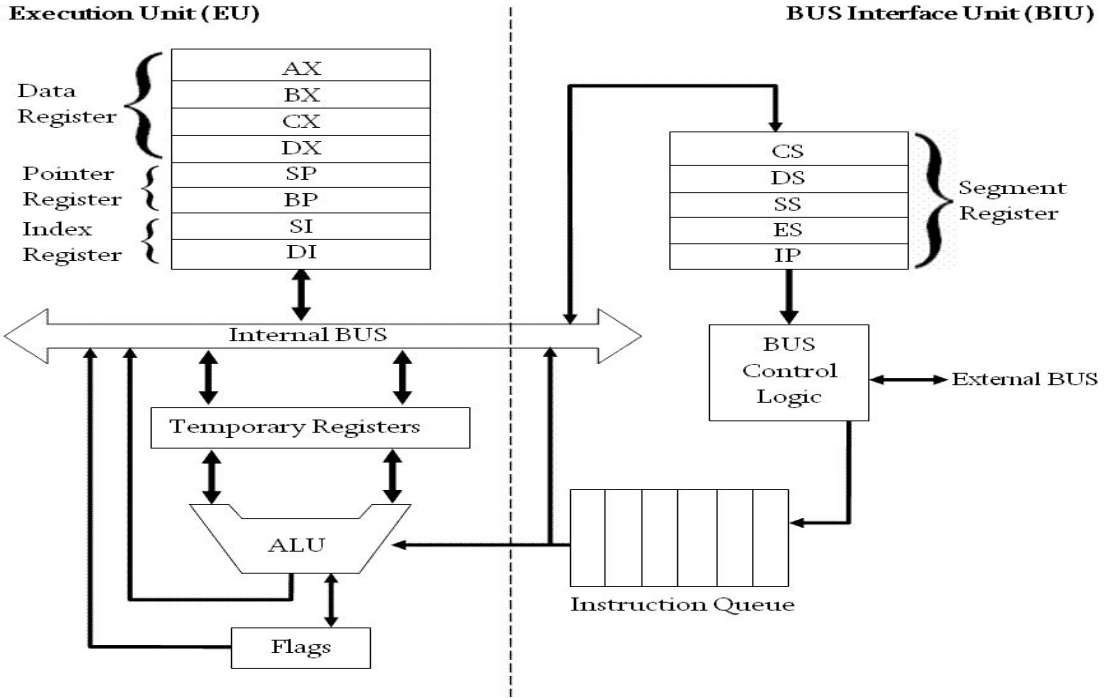
All CPUs are microprocessors.
And 80X86 are **microprocessor chips** designed by Intel.



Von Neumann Diagram

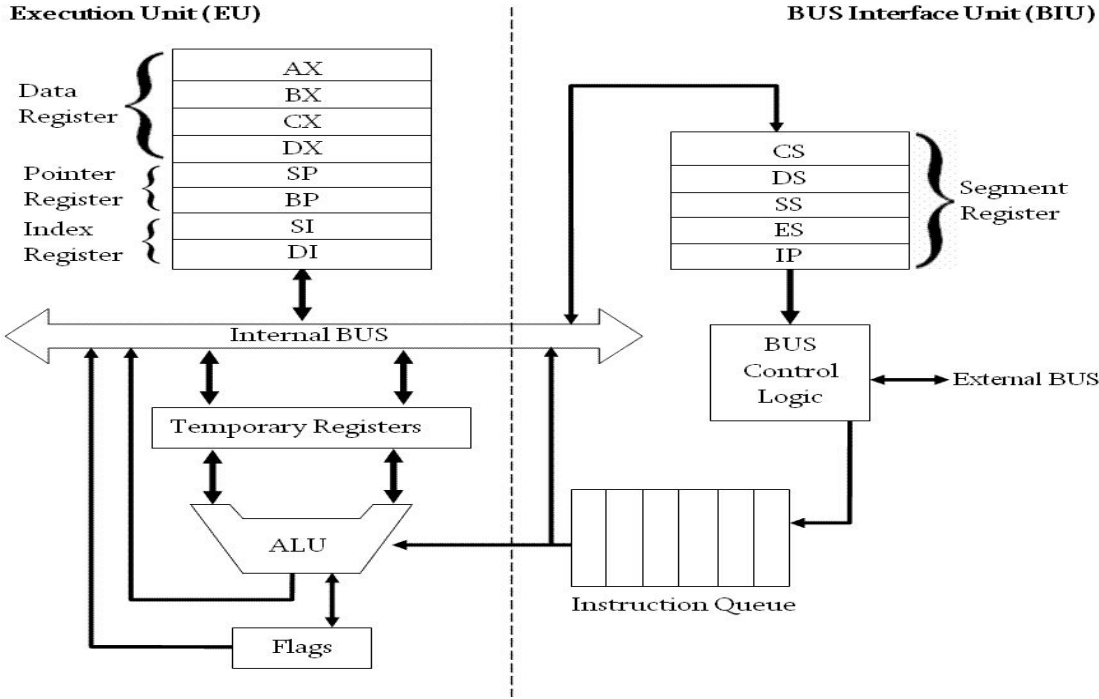
System Architecture: 8086 and Other Components

- 8086 CPU
  - **EU (Execution Unit)**
  - **BIU (Bus Interface Unit)**
- Memory
  - RAM **(Instructions and Data)**
  - ROM (**Firmware**)
- Peripherals (I/O Devices)
  - Keyboard, Display, Disk drivers



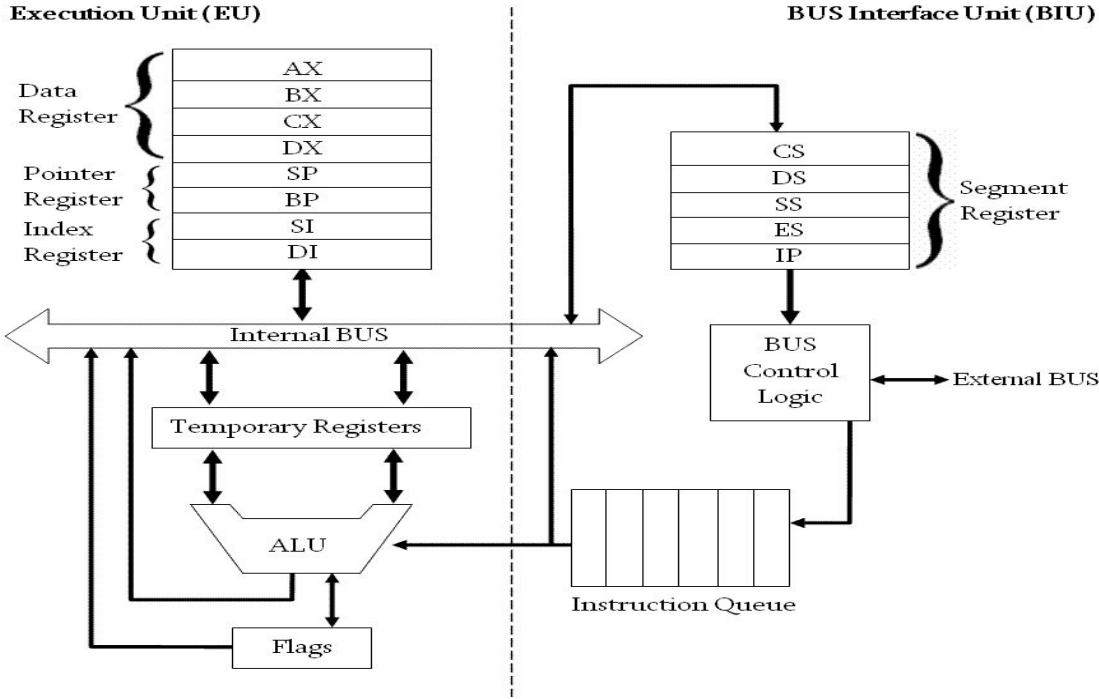
System Architecture: **8086**

- 8086 CPU
  - **EU**
    - **ALU circuit** to perform Arithmetic & Logical Instructions on general regs
    - **General regs** are used by ALU ckt
    - **Flags** is used for control flow or condition
  - **BIU**
    - Use to communicate between **EU** and **RAM** or **I/O**
    - **Have regs** to hold the address of memory location
- 8086 Properties
  - 16-bit **Microprocessor**
    - Can work on 16-bit at a time
  - 20-bit Address **Bus**
    - Can support 2^{20} memory bytes



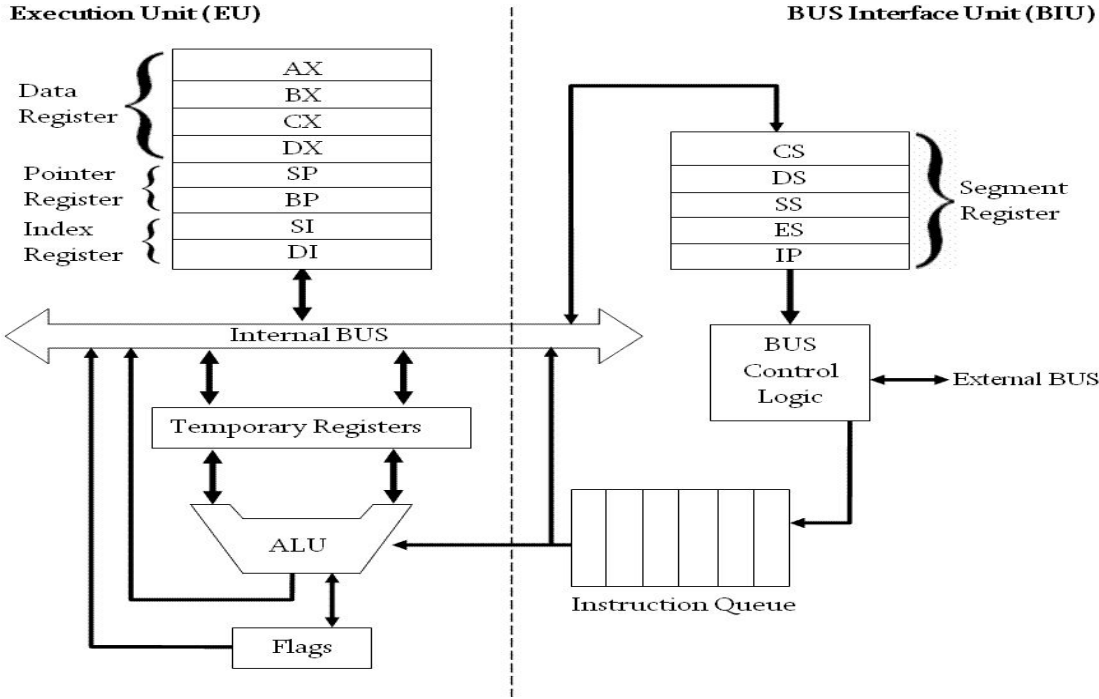
System Architecture: Instruction Execution

- Fetch
  - Bring instructions (Opcode + Operand) from Memory to CPU
- Decode
  - Determine the operation based on Opcode.
- Execute
  - Perform the operation & store the results



Registers

- Data Reg (4)
- Address Reg (9)
- Flag Reg (1)
- Each of them are **16 bit**
- Total 14 register



Registers: Data Registers

- There are 4 general purpose data register.
  - AX (Accumulator Register)
    - Preferred to ALU and Transfer operations as **shortest machine code**
    - Multiplication, Division and I/O
  - BX (Base Register)
    - Address register
  - CX (Count Register)
    - Loop
  - DX (Data Register)
    - Multiplication, Division and I/O

Registers: Address Registers (Segments)

- How programs are stored in 8086?
  - Using Segments
    - Code and Data are stored in different segments
  - Address reg are 16 bit
  - But address bus is 20-bit??
- Logical Address and Physical Address
  - Physical Address (**20 bit**)
  - Logical Address (**Segment: Offset**)
  - Offset are stored in Address regs and Segment are stored in Segment regs
- Four Program Segments in 8086
  - **Code segment**
  - **Data segment**
  - **Stack segment**
  - **Extra Segment**

Registers: Address Registers

- **Segment Register**
  - Hold the address of the segments (**segment number**)
- **Pointer**
  - Point to memory location
- **Index Register**
  - Point to memory location (**Specially DS segment**)

Logical Address to Physical Address

- Logical Address(16-bit) to Physical Address(20-bit)
  - Left shift the segment address 4 bits or Multiplying 10h
  - Then adds the offset
- Example
  - A4FB:4872h → A9822h
  - For the memory location whose physical address is specified for 1256Ah, give the address in **segment:offset** form for segments 1256h and 1240h.
    - **1256Ah = 12560h + X** and **1256Ah = 12400h + Y** → 1256Ah = **1256:000A** = **1240:016A**
  - A memory location has physical address 80FD2h. In what segment does it have offset BFD2h? → 7500h.

Registers: Address Registers

- **Segment Register**
  - CS
  - DS
  - SS
  - ES
- Pointer
  - SP
  - BP
  - **IP**
- Index Register
  - SI
  - DI

Registers: Flag Register

- Indicate the status of the microprocessor
- Two types of flags:
  - Status flags
    - Example: ZF (zero flag)
  - Control flags
    - Example: IF (interrupt flag)

Program Structure

- Statement: Instruction to execute or Directives for the assembler
- Memory Models
  - .MODEL memory_model (SMALL)
- Data Segment
  - Declare variables and constants
- Stack Segment
  - .STACK size (100H)
- Code Segment
  - Define procedures
    - name PROC
    - name ENDP
  - MAIN refers to the main procedure

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE

MAIN PROC

    MAIN ENDP
END MAIN
```

Program Structure

```
.MODEL SMALL
```

This directive tells the assembler which memory model you are using.

SMALL model means:

- Single code segment (max 64 KB)
- Single data segment (max 64 KB)

```
.STACK 100H
```

This defines the size of the stack segment.

- `100H` = 256 bytes (since H means hex).
- The stack is a memory area used for:
  - Function calls (saving return addresses)
  - Pushing/popping registers
  - Local variables (sometimes)

Different types of MODEL

Model	Code Segments	Data Segments	Total Code Size	Total Data Size
TINY	1 (Shared)	1 (Shared)	≤ 64 KB	≤ 64 KB
SMALL	1	1	≤ 64 KB	≤ 64 KB
MEDIUM	Many	1	> 64 KB	≤ 64 KB
COMPACT	1	Many	≤ 64 KB	> 64 KB
LARGE	Many	Many	> 64 KB	> 64 KB
HUGE	Many	Many	> 64 KB	> 64 KB (arrays > 64K)

Program Structure

```
.DATA
```

This starts the data segment of your program.

- Here, you'd declare variables, strings, or arrays.
- In this code, it's empty for now.

Example:

.DATA

msg DB "Hello, World!$"

num DW 1234

```
.CODE
```

This begins the code segment—the instructions your CPU will execute.

Program Structure

```
MAIN PROC

    MAIN ENDP
```

This declares the start and end of a procedure(function) named MAIN.

```
END MAIN
```

This is where the program ends.

But why need END MAIN when we are already using MAIN ENDP?

Emulator

- We will use emulator to see how the 8086 execute instructions line by line.

Link: Emu 8086
Online Emulator: Emu8086 Online

Any Question?





Assembly Language

Registers

- Type of a fast memory
- Used to store and transfer data and instructions

Types of Registers

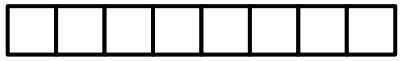
- 3 types
  - Data Register
  - Address Register
  - Status Register

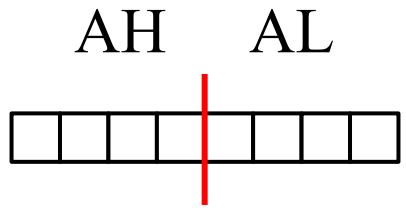
We will mainly focus on ***Data Register***

Data Register

- Holds data
- 16 bit
- 4 types:
  - AX: Accumulator Register
  - BX: Base Register
  - CX: Count Register
  - DX: Data Register

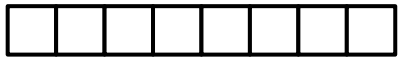
AX: Accumulator Register

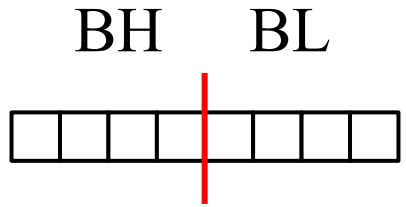
- 16 bit AX: 
- Upper and lower bytes can be used separately



- Special Purpose: Multiplication, Division, Input, Output

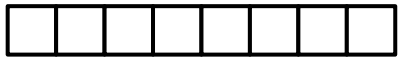
BX: Base Register

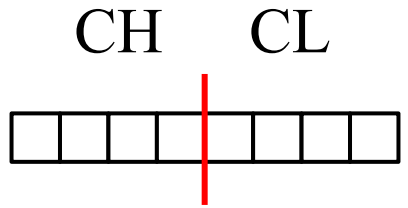
- 16 bit BX: 
- Upper and lower bytes can be used separately



- Special Purpose: General purpose register

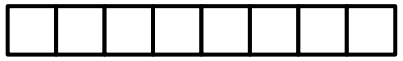
CX: Count Register

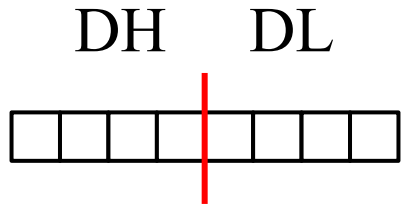
- 16 bit CX: 
- Upper and lower bytes can be used separately



- Special Purpose: Controlling loop

DX: Data Register

- 16 bit DX: 
- Upper and lower bytes can be used separately



- Special Purpose: Multiplication, Division, Output

Representing Numbers

- Binary Numbers
  - Must end with ***b***
  - Example: 10110111b
- Hexa Numbers
  - Must end with ***h***
  - Must start with ***digit***
  - Legal Number: 2Ah
  - Illegal Number: DAh
  - Legal Number: 0DAh
- Decimal Numbers
  - Can end with ***d***
  - Legal Number: 128d
  - Legal Number: 128
- Find the base of the following numbers
  - 10110111b **Binary**
  - 1Bh **Hexa**
  - 10110111 **Decimal**
  - 3B **Wrong Format**
  - 38 **Decimal**
  - EAh **Wrong Format** **Correct: 0EAh**

Naming of Variables

- Can be combination of
  - Letter
  - Digit
  - Special Characters like: @ $? _
- Can not be started with digit and can not be consist of only digits
- Case insensitive
- Verify the following variable names:
  - 1Var db 127 **Illegal**
  - 123 db 127 **Illegal**
  - ? db 127 **Legal**
  - AbC$? db 127 **Legal**
  - @1 db 127 **Legal**

Registers and Data Types

Register	*16 bit*	*8 bit*
Accumulator	AX	AH, AL
Base	BX	BH, BL
Count	CX	CH, CL
Data	DX	DH, DL

- List of Registers

Data type	*Full form*	*Number of bits*
DB	Define Byte	8
DW	Define Word	16
DD	Define Double Word	32
DQ	Quad Word	64

- List of Data Types

Declaration of Variables

Variable Name	Data type	Initial Value
var1	db	5

- Must be initialized with a legal value
- Verify the following initializations:
  - Var1 db 320 **Illegal**
  - Var1 db 10011001101O1b **Illegal**
  - Var1 db 00000001101O1b **Legal**

Format of Instruction

Opcode	Operands	Comments
Mov	CX,4	; We don't code in C, we code in Assembly

Commonly Used Opcode

MOV

- MOV DEST, SRC
- DEST = Registers, Variables
- Example: MOV CX, 8
- Equivalent Task: DEST = SRC
- SRC = Registers, Variables, Values
- Equivalent Task: CX = 8
- Number of bits (SRC) = Number of bits (DEST)
- SRC and DEST can not be variables at a time
- Verify the following instructions:
  - MOV AH, DL **;Legal** ; AH = DL
  - MOV var1, AL **;Legal if Var1 is DB** ; var1 = AL
  - MOV AX, Var1 **;Legal if Var1 is DW** ; AX = var1
  - MOV var1, var2 **;Illegal**
  - MOV AL,000000000011b **;Legal**
  - MOV CL, 100000000000b **;Illegal**
  - MOV BH, 287 **;Illegal**
  - MOV var1, 125 **;Legal**

XCHG

- XCHG ARG1, ARG2
- ARG1 = Registers, Variables
- Example: XCHG AL, BH
- Equivalent Task: SWAP(ARG1, ARG2)
- ARG2 = Registers, Variables
- Equivalent Task: SWAP(AL, BH)
- Number of bits (ARG1) = Number of bits (ARG2)
- ARG1 and ARG2 can not be variables at a time
- Verify the following instructions:
  - XCHG AH, DL **;Legal** ; SWAP(AH, DL)
  - XCHG var1, AL **;Legal if Var1 is DB** ; SWAP(var1, AL)
  - XCHG AX,30 **;Illegal**
  - XCHG var1, var2 **;Illegal**

ADD

- ADD DEST, SRC
- DEST = Registers, Variables
- Example: ADD AL, BH
- Equivalent Task: DEST= DEST + SRC
- SRC = Registers, Variables, Values
- Equivalent Task: AL = AL + BH
- Number of bits (SRC) = Number of bits (DEST)
- SRC and DEST can not be variables at a time
- Verify the following instructions:
  - ADD AH, DL **;Legal** ; AH=AH+DL
  - ADD var1, AL **;Legal if Var1 is DB** ; var1=var1+AL
  - ADD AX, Var1 **;Legal if Var1 is DW** ; AX=AX+var1
  - ADD var1, var2 **;Illegal**
  - ADD var1, 125 **;Legal** ; var1=var1+125
  - ADD ch, 125 **;Legal** ; ch=ch+125
  - ADD 320, al **;Illegal**

SUB

- SUB DEST, SRC
- DEST = Registers, Variables
- Example: SUB AL, BH
- Equivalent Task: DEST= DEST - SRC
- SRC = Registers, Variables, Values
- Equivalent Task: AL = AL - BH
- Number of bits (SRC) = Number of bits (DEST)
- SRC and DEST can not be variables at a time

NEG

- NEG DEST
- DEST = Registers, Variables
- Example: NEG AL

- Equivalent Task: DEST= - DEST
- Equivalent Task: AL = - AL

INC

- INC DEST
- DEST = Registers, Variables
- Example: INC AL

- Equivalent Task: DEST= DEST + 1
- Equivalent Task: AL = AL + 1

DEC

- DEC DEST
- DEST = Registers, Variables
- Example: DEC AL

- Equivalent Task: DEST= DEST - 1
- Equivalent Task: AL = AL - 1

Task

- Write instructions in Assembly Language for the following task:
  - A = B – 2*A [Consider A and B as db type variables]

```
MOV  AL, A      ; AL = A
ADD  AL, AL     ; AL = 2*A
SUB  AL, B      ; AL = 2*A - B
MOV  A, AL      ; A = 2*A - B
NEG  A          ; A = B – 2*A
```

Single Character Input

- Required Instruction: INT 21H
- Set AH = 1 before calling INT 21H
- Input character will be stored at ***AL*** after calling INT 21H
- Value of AH works as Mode Selector here

```
MOV  AH,1
INT  21H
; Now use the input from AL
```

Single Character Output

- Required Instruction: INT 21H
- Set AH = 2 before calling INT 21H
- Store the output character at ***DL*** before calling INT 21H
- Value of AH works as Mode Selector here

```
; Set your required character at DL
MOV  AH,2
INT  21H
```

String Output

- Required Instruction: INT 21H
- Set AH = 9 before calling INT 21H
- Store the output string with a $ at the end as a DB type variable
- Move the variable at DX by LEA before calling INT 21H //LEA= Load Effective Address

```
STRING1  DB  'HELLO WORLD$'
MOV  AH,9
LEA  DX, STRING1
INT  21H
```

String Output

```
.MODEL SMALL
.STACK 100H

.DATA
STRING1 DB "HELLO CSE23$"

.CODE

MAIN PROC

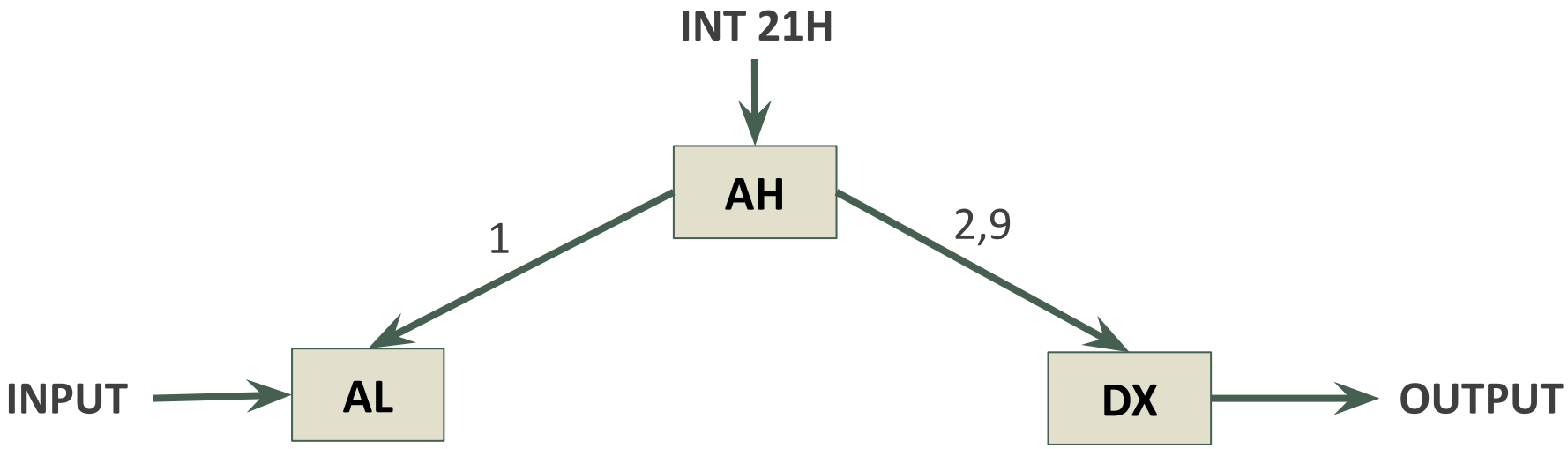
MOV AX, @DATA
MOV DS, AX

MOV AH,9
LEA DX,STRING1
INT 21H

MOV AH, 4Ch
INT 21H

MAIN ENDP
END MAIN
```

Summary



Point To Remember

- Value of AL gets changed after calling INT 21H
- So, if you have a desired value at AL, store it at other place before calling INT 21H



Thank You

QUESTION?

Lecture-03

Flow Control

- Unconditional Jump: JMP
- Conditional Jump: JE, JNE, JG, JGE, JL, JLE etc.
- Loop

Unconditional Jump:

- **JMP Syntax**

JMP Label

…..

…..

Label:

Special Instruction : CMP

CMP **dest**, **src**

- **dest** can be a register or a variable
- **src** can be a register, a variable or a value
- Subtracts dest value from src value and updates the flag register accordingly

Example:

```
CMP AX,0000H
```

Conditional Jump

JE => Jump if Equal

Label:

…..

…..

CMP AX,0000H

JE **Label**

Conditional Jump

JNE => Jump if not Equal

Label:

…..

…..

CMP AX,0000H

JNE **Label**

Conditional Jump

JG => Jump if Greater

JGE => Jump if Greater or Equal

JL => Jump if Less

JLE => Jump if Less or Equal

Conditional Jump

JXCZ => Jump if CX is Zero

JC => Jump if Carry

JNC => Jump if No Carry

JO => Jump if Overflow

JNO => Jump if No Overflow

JP => Jump if Parity

JNP => Jump if No Parity

JS => Jump if Sign

JNS => Jump if No Sign

Practice Problems

1. Write a Assembly program to find maximum between two numbers.
2. Write a Assembly program to find maximum between three numbers.
3. Write a Assembly program to check whether a number is negative, positive or zero.
4. Write a Assembly program to check whether a number is even or odd.



Flow Control: Loop

Loop

- Basic Syntax:

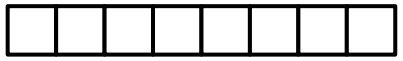
Label : ……

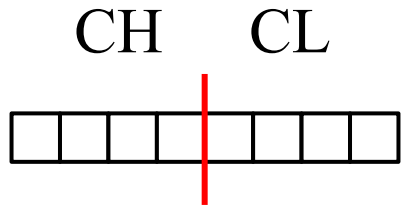
……

……

Loop **Label**

Loop : Count Register

- 16 bit CX: 
- Upper and lower bytes can be used separately



- Special Purpose: Controlling loop

Loop

- Decrements the value of CX register by 1 after every iteration
- Checks the value of CX register after each iteration
- Jumps to the specified label or break out of the loop depending on the value in CX register

- How to output multiple characters one by one in assembly?

  =>The answer is **Loop**

Practice Problems

- Print from A to Z sequentially using loop in assembly language.
- Print the following pattern when input is 3

```
*
**
***
```



Thank You

QUESTION?



ASSEMBLY LANGUAGE

Logical Operations

- AND
- OR
- TEST
- XOR
- SHL
- SHR
- ROL
- ROR
- RCL
- RCR

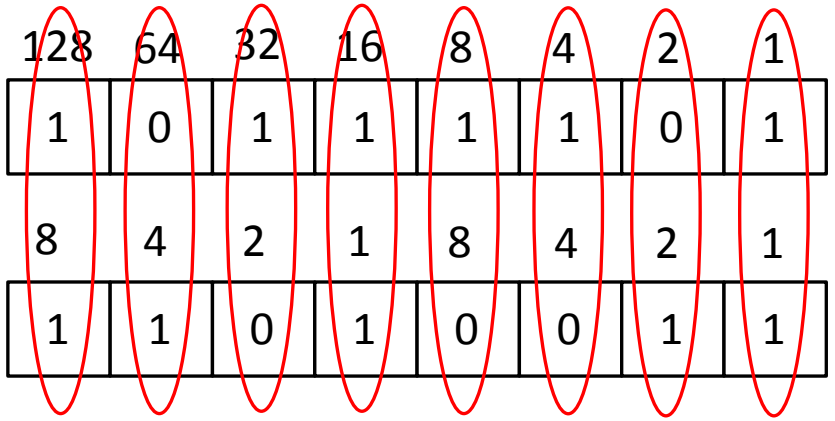
AND

- Performs bitwise AND operation
- Instruction format: AND DEST, SRC
- Example: AND AL, 3
- Equivalent operation: DEST = DEST AND SRC
- Equivalent operation: AL = AL AND 3

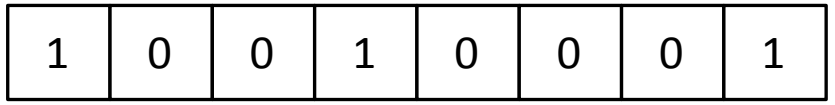
- *Restrictions*:
- Number of bits (SRC) = Number of bits (DEST)
- SRC and DEST can not be variables at a time

AND

- *Working Principle*:
  - Let AL = 189d
  - Let BL = 0D3h



- AL AND BL



- AL in decimal 145d
- AL in Hexa 91h

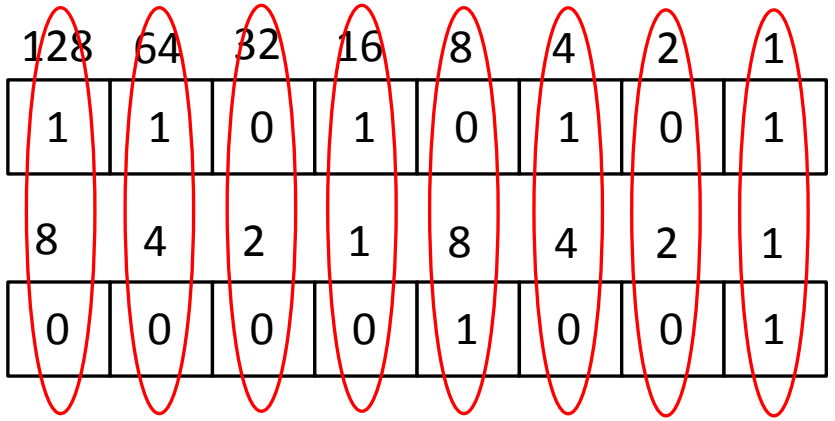
OR

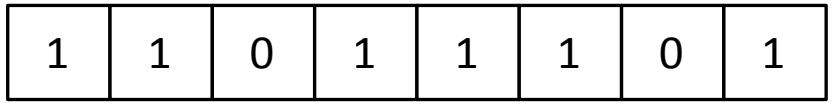
- Performs bitwise OR operation
- Instruction format: OR DEST, SRC
- Example: OR AL, 3
- Equivalent operation: DEST = DEST OR SRC
- Equivalent operation: AL = AL OR 3

- *Restrictions*:
- Number of bits (SRC) = Number of bits (DEST)
- SRC and DEST can not be variables at a time

OR

- *Working Principle*:
  - Let AL = 213d
  - Let BL = 9h





- AL OR BL
- AL in decimal 221d
- AL in Hexa 0DDh

XOR

- Performs bitwise XOR operation
- Instruction format: XOR DEST, SRC
- Example: XOR AL, 3
- Equivalent operation: DEST = DEST XOR SRC
- Equivalent operation: AL = AL XOR 3

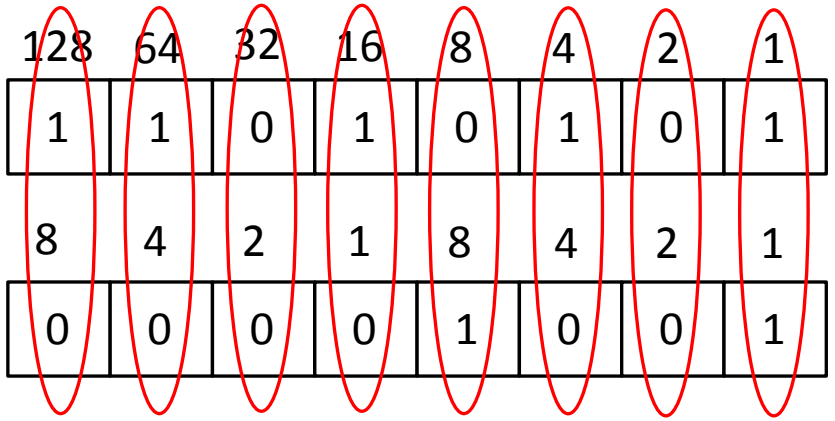
- *Restrictions*:
- Number of bits (SRC) = Number of bits (DEST)
- SRC and DEST can not be variables at a time

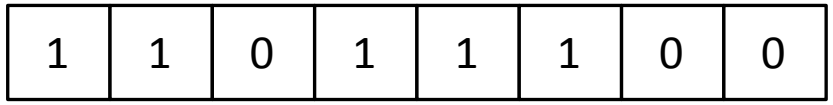
- *Special Properties of XOR*:
- X XOR 0 = X
- X XOR 1 = $\overline{X}$

X	Y	F
0	0	0
1	0	1
0	1	1
1	1	0

XOR

- *Working Principle*:
  - Let AL = 213d
  - Let BL = 9h





- AL XOR BL
- AL in decimal 220d
- AL in Hexa 0DCh

SET A BIT

- Consider AL:
- Bitwise OR with:

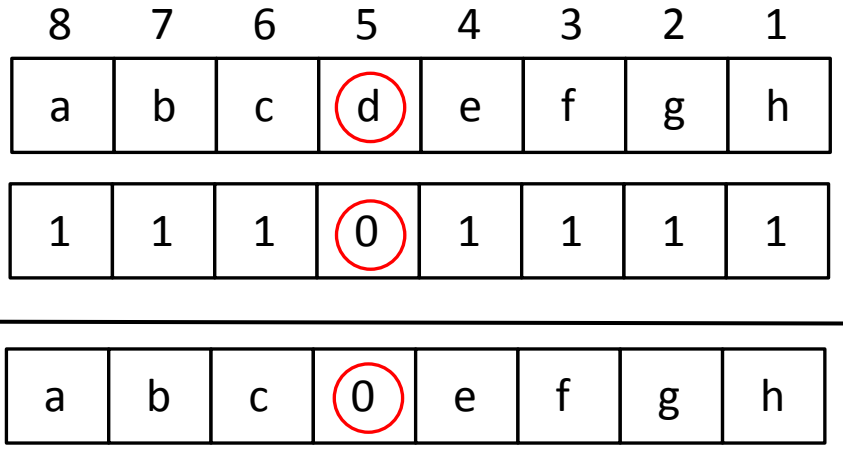
8	7	6	5	4	3	2	1
a	b	c	d	e	f	g	h
0	0	0	0	1	0	0	0

a	b	c	d	1	f	g	h

- Set the 4th bit of AL to 1 and keep other bits unchanged

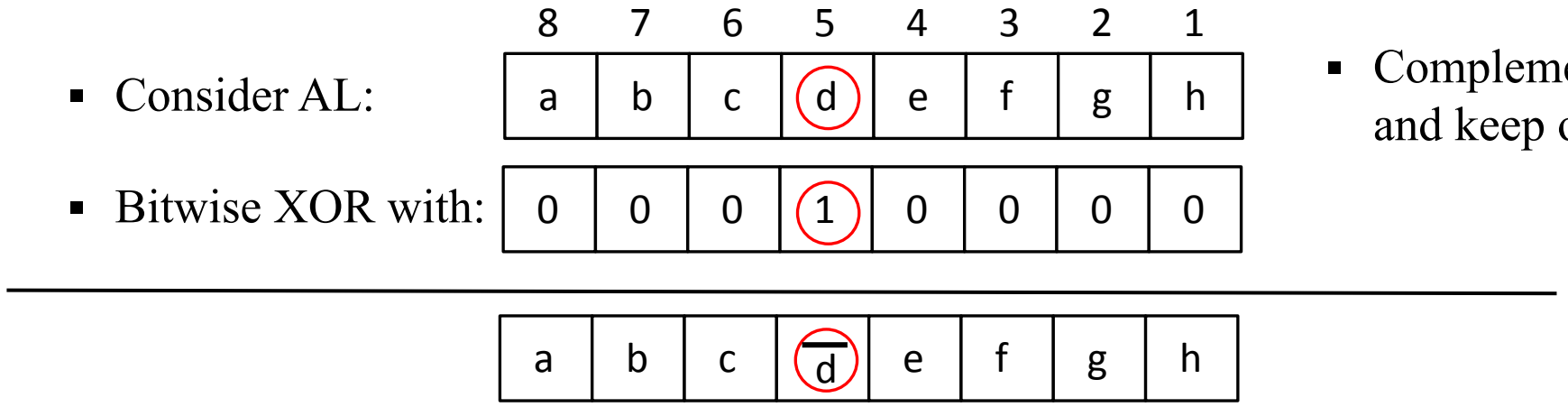
RESET A BIT

- Consider AL:
- Bitwise AND with:



- Set the 5th bit of AL to 0 and keep other bits unchanged

COMPLEMENT/CHANGE A BIT



- Complement the 5th bit of AL and keep other bits unchanged

COMPLEMENT A REGISTER

- Consider AL:

8	7	6	5	4	3	2	1
a	b	c	d	e	f	g	h

- Bitwise XOR with:

1	1	1	1	1	1	1	1

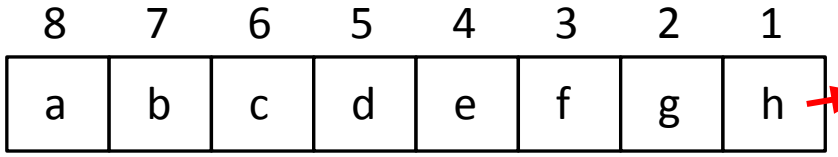
$\overline{a}$	$\overline{b}$	$\overline{c}$	$\overline{d}$	$\overline{e}$	$\overline{f}$	$\overline{g}$	$\overline{h}$

- Complement all the bits of AL

CHECK ODD/EVEN

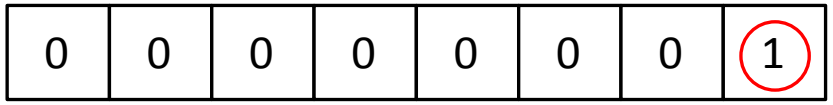
- Check whether the value stored in AL is even or odd

- Consider AL:

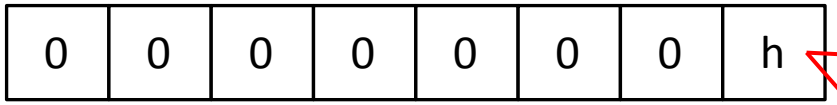


- h is 0 if AL is even, h is 1 if AL is odd

- Bitwise AND with:



- AL:



- If h=1, then AL=1, means AL is odd
- If h=0, then AL=0, means AL is even

```
CMP AL,1
JE ODD
JNE EVEN
```

TEST

- Performs bitwise AND operation
- Instruction format: TEST A, B
- Difference with AND is it doesn't store the resultant value
- But it changes the value of Zero Flag
- If the resultant value is 0, it sets the Zero Flag to 1, else 0
- Equivalent operation: TEST A,B means if A AND B is 0 then Z=1, else Z=0
- *Restrictions*:
- Number of bits (A) = Number of bits (B)
- A and B can not be variables at a time

TEST EXAMPLE

- Consider A:

8	7	6	5	4	3	2	1
0	1	1	0	1	1	0	1

- Consider B:

1	0	0	0	0	0	1	0

- A AND B:

0	0	0	0	0	0	0	0

- Result is 0
- So, Z will be set to 1

TEST EXAMPLE

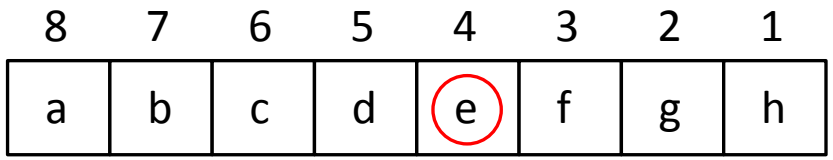


	8	7	6	5	4	3	2	1
▪ Consider A:	0	1	1	0	1	1	0	1
▪ Consider B:	1	1	0	0	0	0	1	1
▪ A AND B:	0	1	0	0	0	0	0	1

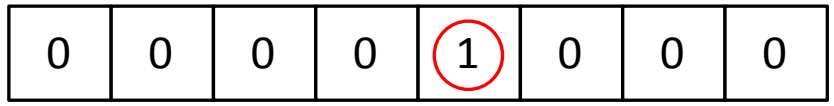
- Result is 65 (Not Zero)
- So, Z will be set to 0

CHECK A BIT

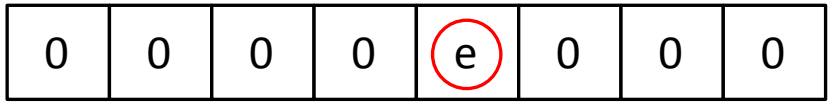
- Consider AL:



- TEST AL with:



- TEST result:



- Check whether the 4th bit of AL is 0 or 1
- If 4th bit (e) is 1, then Test result is not Zero, means Z=0
- If 4th bit (e) is 0, then Test result is Zero, means Z=1

```
TEST AL,00001000b
JZ Fourth_Bit_Is_Zero
JNZ Fourth_Bit_Is_One
```

CHECK ODD/EVEN

- Check whether the value stored in AL is even or odd

	8	7	6	5	4	3	2	1
Consider AL:	a	b	c	d	e	f	g	h
TEST AL with:	0	0	0	0	0	0	0	1
TEST Result:	0	0	0	0	0	0	0	h

```
TEST AL,1
JZ EVEN
JNZ ODD
```

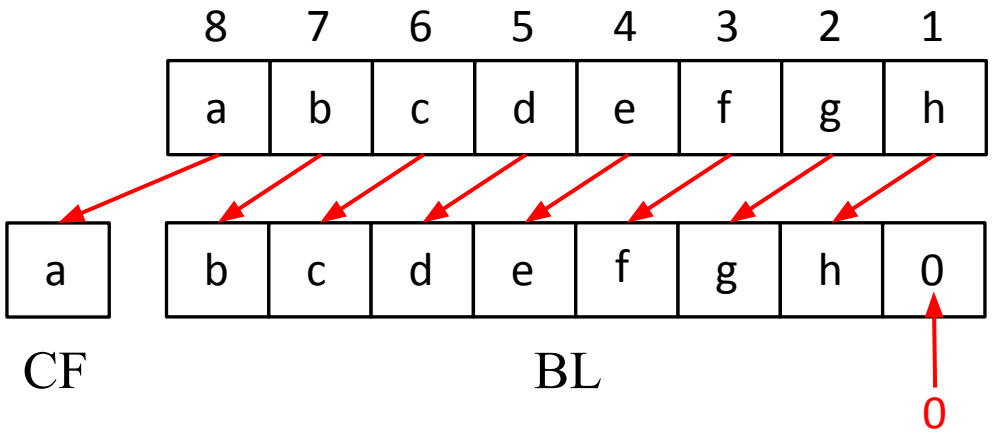
SHIFT LEFT

- Performs bitwise Left Shift operation
- Instruction format: SHL A, B
- Shifts A for B number of times at left
- Stores the MSB at Carry Flag (CF) after one left shift

- *Restrictions*:
- ***B must be value or CL***

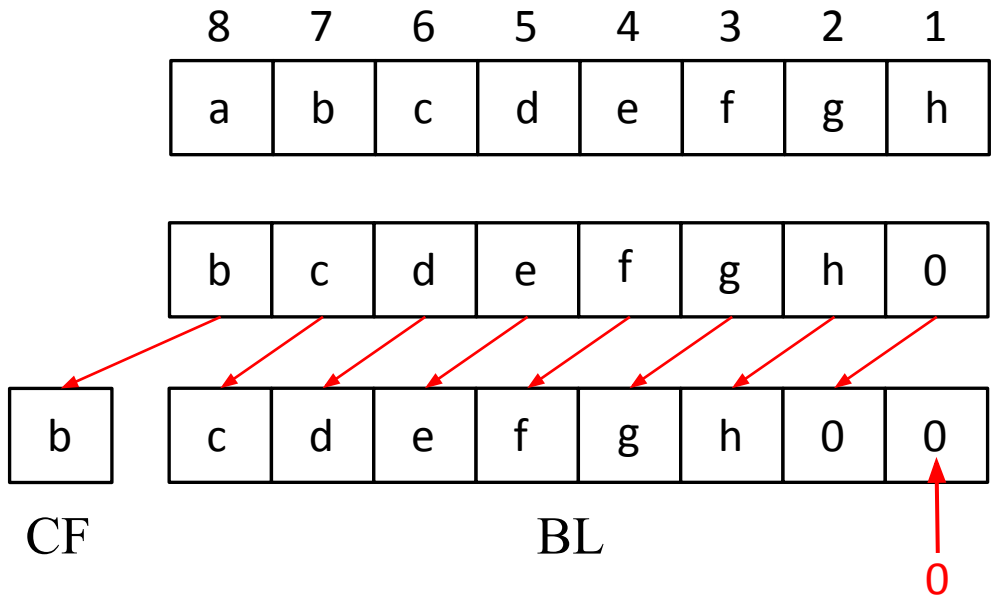
SHIFT LEFT

- Consider BL:
- SHL BL,1:



SHIFT LEFT

- Consider BL:
- SHL BL,1:
- Again SHL BL,1:



SHIFT LEFT

- Maximum how many Left Shift operation needed for setting a n-bit register to 0?
  - n
- Write a code that sets BL to 0 by Left Shift operation

First Way:

```
SHL BL,8
```

Second Way:

```
MOV CL,8
SHL BL,CL
```

Wrong Way:

```
MOV AL,8
SHL BL,AL
```

- Write a code for printing the binary value of a 8 bit register/variable

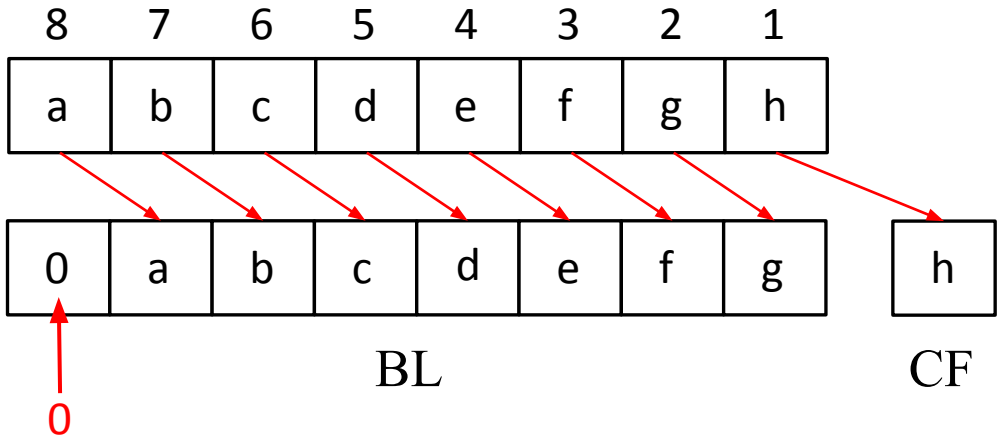
SHIFT RIGHT

- Performs bitwise Right Shift operation
- Instruction format: SHR A, B
- Shifts A for B number of times at right
- Stores the LSB at Carry Flag (CF) after one right shift

- *Restrictions*:
- ***B must be value or CL***

SHIFT RIGHT

- Consider BL:
- SHR BL,1:



ROTATE LEFT

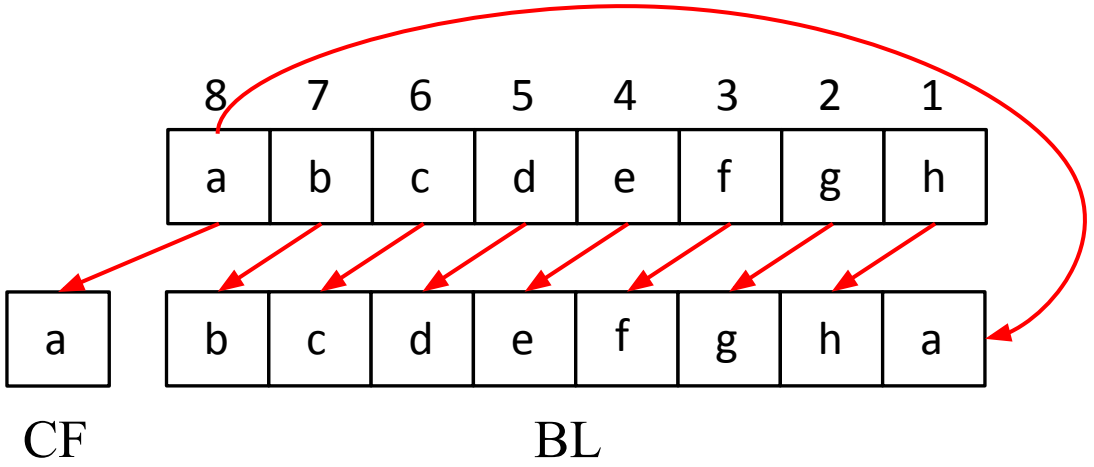


- Performs bitwise Left Rotate operation
- Instruction format: ROL A, B
- Rotate A for B number of times at left
- Stores the MSB at Carry Flag (CF) and at LSB after one left rotate

- *Restrictions*:
- ***B must be value or CL***

ROTATE LEFT

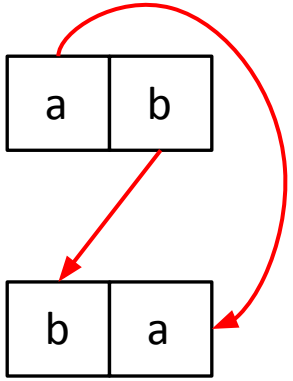
- Consider BL:
- ROL BL,1:



ROTATE LEFT

- Let X is a 2 bit register, Rotate it for 2 times (left/right), And then write down the observation

- A:

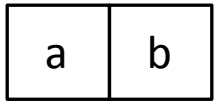


- ROL AL,1:

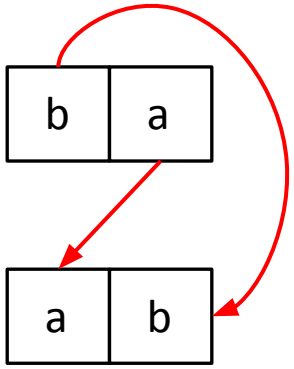
ROTATE LEFT

- Let X is a 2 bit register, Rotate it for 2 times (left/right), And then write down the observation

- A:



- ROL AL,1:



- ROL AL,1:

- The initial value gets restored if a n-bit register/memory is rotated(left/right) for n times

- Write down the advantages of printing the binary value of a register/variable using rotate operation rather than using shift operation.

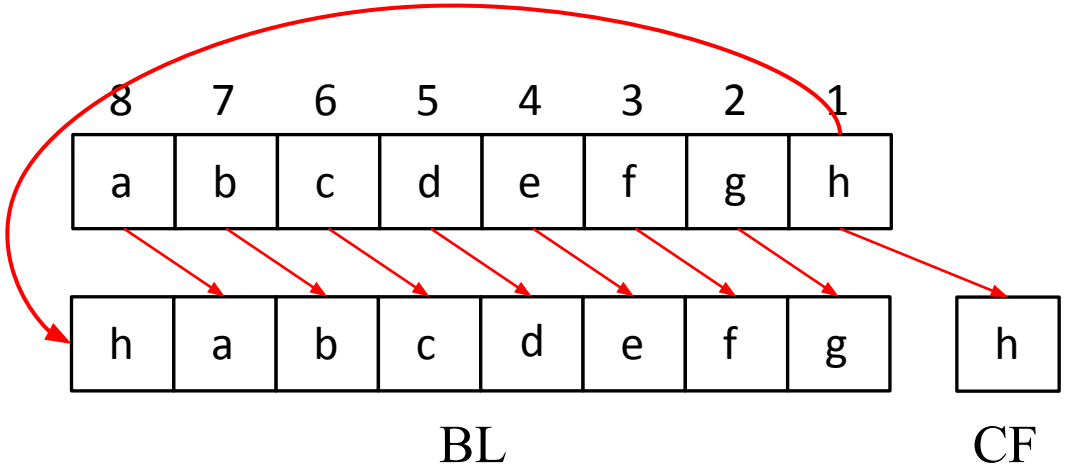
ROTATE RIGHT

- Performs bitwise Right Rotate operation
- Instruction format: ROR A, B
- Rotate A for B number of times at right
- Stores the LSB at Carry Flag (CF) and at MSB after one left rotate

- *Restrictions*:
- ***B must be value or CL***

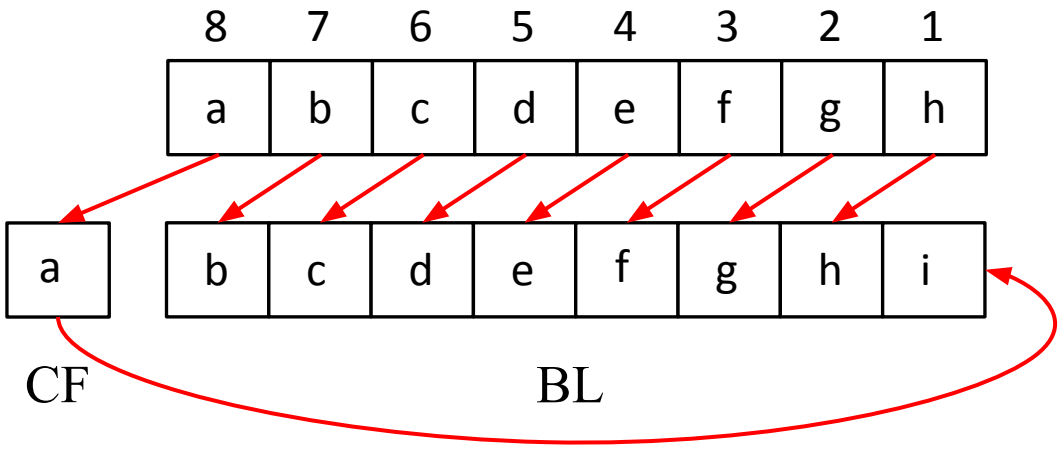
SHIFT RIGHT

- Consider BL:
- SHR BL,1:



ROTATE CARRY LEFT

- Consider BL:
- RCL BL,1:



- ROTATE CARRY RIGHT will be in same way

QUESTIONs

- We XOR 32 with a letter for converting it from uppercase to lowercase or lowercase to uppercase. Why?
- One time Left shift of a register/variable results into a multiplication by 2, do you agree?
- **n** time Left shift of a register/variable results into a multiplication by 2^n, do you agree? [n<8]
- Set the value of Al register as 0, using only XOR operation. MOV can not be used

QUESTIONs

- Only difference between Code-1 and Code-2 is that sequence of writing the labels “print_one” and “print_zero” are swapped. But Code-1 will execute for infinite times and Code-2 will execute as expected (8 times). Explain this fact.

```
mov bl, 10110001b
mov cx,8
mov ah,2; for output

BEGIN:
rol bl,1
jc print_one
Jnc print_zero

print_one:
mov dl,'1'
int 21h
loop BEGIN

print_zero:
mov dl,'0'
int 21h
loop BEGIN
```

Code-1

```
mov bl, 10110001b
mov cx,8
mov ah,2; for output

BEGIN:
rol bl,1
jc print_one
Jnc print_zero

print_zero:
mov dl,'0'
int 21h
loop BEGIN

print_one:
mov dl,'1'
int 21h
loop BEGIN
```

Code-2



Number Input Output

Binary Input-Output

Algorithm for Binary Input

- For binary Input, we assume a program reads in a binary number from the keyboard, followed by a **carriage return**.
- The number actually is a character string of 0's and 1's.
- As each character is entered, we need to convert it to a bit value, and insert the bit in a register.

Algorithm for Binary Input

Suppose Input: 1010

					Input
Before Itr 1	0	0	0	0	1
Itr 1	0	0	0	1	0
Itr 2	0	0	1	0	1
Itr 3	0	1	0	1	0
Itr 4	1	0	1	0	CR

Algorithm for Binary Input

```
Clear BX /* BX will hold binary value */
Input a character /* '0' or '1' */
WHILE character <> CR DO
    Convert character to binary value
    Left shift BX
    Insert value into LSB of BX
    Input a character
END_WHILE
```

Algorithm for Binary Output

Suppose Output: 1010

					CF
Before Itr	1	0	1	0	X
Itr 1	0	1	0	1	1
Itr 2	1	0	1	0	0
Itr 3	0	1	0	1	1
Itr 4	1	0	1	0	0

Algorithm for Binary Output

```
FOR 16 times DO
    Rotate left BX /* BX holds output value, put msb into CF
    */
    IF CF = 1
    THEN
        Output '1'
    ELSE
        Output ‘0’
    END_IF
END_FOR
```

Code of Binary Input/Output

https://ideone.com/pLDGCZ

Hex Input-Output

Algorithm for Hex Input

- Hex input consists of digits ("0" to "9") and letters ("A" to "F'”) followed by a carriage return. For simplicity, we assume that:
  - only uppercase letters are used, and
  - the user inputs no more than four hex characters.
- Unlike Binary input, need to shift the register four times.

Algorithm for Hex Input

```
Clear BX /* BX will hold input value */
Input a hex character /* '0' tO '9' or 'A' tO 'F'*/
WHILE character <> CR DO
    Convert character to binary value
    Left shift BX 4 times
    Insert value into lower 4 bits of BX
    Input a character
END_WHlLE
```

Algorithm for Hex Output

```
FOR 4 times DO
    Mov BH to DL /* "BX holds output value */
    Shift DL 4 times to the right
    IF DL < 10
    THEN
        Convert to character in '0' to '9
    ELSE
        Convert to-character in 'A' to 'F'
    END_IF
    Output character
    Rotate BX left 4 times
END FOR
```



Thank You

QUESTION?



Multiplication/Division

Objectives

- Multiplication Instructions
- Division Instructions
- Decimal I/O

Book

Multiplication Instructions

- Syntax: `Opcode Source`
- `Opcode`
  - "MUL" for unsigned multiplication
  - "IMUL" for signed multiplication
- `Source`
  - Byte (8 bit) => `AX = AL * Source`
  - Word (16 bit) => `DX:AX = AX * Source`

Effect of MUL/ IMUL on the status flags	
SF, ZF, AF, PF	Undefined
CF/ OF:	
After MUL	= 0 if the upper half of the result is zero.
	= 1 otherwise.
After IMUL	= 0 if the upper half of the result is the sign extension of the lower half (this means that the bits of the upper half are the same as the sign bit of the lower half).
	= 1 otherwise.

- If the lower destination(AL or AX) can not store the result then CF/OF is 1.

Multiplication Instructions: Examples

1. Suppose AX contains 1 and BX contains FFFFh. → MUL BX, IMUL BX
2. Suppose AX contains FFFFh and BX contains FFFFh. → MUL BX, IMUL BX
3. Suppose AX contains 0FFFh. → MUL AX, IMUL AX
4. Suppose AX contains 0100h and CX contains FFFFh. → MUL CX, IMUL CX
5. Suppose AL contains 80h and BL contains FFh. → MUL BL, IMUL BL

What will be the value of the registers, CF and OF Flags after the **MUL** and **IMUL** instructions.

Class Assessment

Convert the following High Level Instruction to Assembly Language using IMUL (suppose there is no overflow):

A = 5 x A - 12 x B

Class Assessment

Convert the following High Level Instruction to Assembly Language using IMUL (suppose there is no overflow):

A = 5 x A - 12 x B

```
MOV AX, 5       ;AX m 5
IMUL A          ;AX = 5 x A
MOV A, AX       ;A = 5 x A
MOV AX, 12      ;AX = 12
IMUL B          ;AX = 12 x B
SUB A, AX       ;A = 5 x A - 12 x B
```

MUL Operation: Application

Algorithm for Factorial.

```
FACTORIAL PROC
;computes N! ·
;input: CX = N
;output: AX = N!
    MOV AX,1
TOP:·
    MUL CX
    LOOP TOP
    RET
FACTORIAL ENDP
```

Division Instructions

- Syntax: **`Opcode Divisor`**
- **`Opcode`**
  - "DIV" for unsigned division
  - "IDIV" for signed division
- **`Divisor`**
  - Byte (8 bit) => `AL = AX / Divisor` **and** `AH = AX % Divisor`
  - Word (16 bit) => `AX = DX:AX / Divisor` **and** `DX = DX:AX % Divisor`

Division Instructions: Examples

1. Suppose DX contains 0000h, AX contains 0005h, and BX contains 0002h. → DIV BX, IDIV BX
2. Suppose DX contains 0000h, AX contains 0005h, and BX contains FFFEh. → DIV BX, IDIV BX
3. Suppose DX contains FFFFh, AX contains FFFBh, and BX contains 0002h. → DIV BX, IDIV BX
4. Suppose AX contains 00FBh and BL contains FFh. → DIV BL, IDIV BL

What will be the value of the registers after the **DIV** and **IDIV** instructions.

Sign Extension of the Dividend

Word Division	Byte Division
For DIV, DX should be cleared.	For DIV, AH should be cleared.
For IDIV, DX should be made the sign extension of AX. The instruction **CWD** (convert word to doubleword) will do the extension.	For IDIV, AH should be made the sign extension of AL. The instruction **CBW** (convert byte to word) will do the extension.

Class Assessment

Convert the following High Level Instruction to Assembly Language using IMUL (suppose there is no overflow):

A = -1250 / B

Class Assessment

Convert the following High Level Instruction to Assembly Language using IMUL (suppose there is no overflow):

A = -1250 / B

```
MOV AX,-1250    ;AX gets dividend
CWD             ;Extend sign to DX
MOV BX,7        ;BX has divisor
IDIV BX         ;AX gets quotient, DX has remainder
```

Decimal Input: Algorithm

```
total = 0
read an ASCII digit
REPEAT
    convert character to a value
    total = 10 x total + value
    read a character
UNTIL character is a carriage return
```

Decimal Input: Code

https://ideone.com/ZWaxc8

Decimal Input

Input: 123

Initialize, total = 0			
Iter=0	read ‘1’	convert ‘1’ to 1	total = 10 x 0 + 1 = 1
Iter=1	read ‘2’	convert ‘2’ to 2	total = 10 x 1 + 2 = 12
Iter=2	read ‘3’	convert ‘3’ to 3	total = 10 x 12 + 3 = 123

Decimal Output: Algorithm

```
IF AX < 0 /• AX holds output value •/
THEN
    print a minus sign
    replace AX by its two's complement
END IF
Get the digits in AX's decimal representation
Convert these digits to characters and print Them
```

Decimal Output: Algorithm

```
Get the digits in AX' s decimal representation
count = 0 /• will count decimal digits •/
REPEAT
     divide quotient by lO
     push remainder on the stack
     count = count + l
UNTIL quotient = 0
```

```
Convert these digits to characters and print Them
FOR count times DO
     pop a digit from the stack
     convert it to a character
     output the character
END_FOR
```

Decimal Output: Code

https://ideone.com/R2ib1w

Decimal Output

Output: 24618

Initially, AX = 24618			
Iter=0	Divide 24618 by 10	Quotient= 2461	remainder= 8
Iter=1	Divide 2461 by 10	Quotient= 246	remainder= 1
Iter=2	Divide 246 by 10	Quotient= 24	remainder= 6
Iter=3	Divide 24 by 10	Quotient= 2	remainder= 4
Iter=4	Divide 2 by 10	Quotient= 0	remainder= 2

Practice Problems

1. Write an assembly code to input two unsigned and output the product
2. Write an assembly code to input two signed number and output the product
3. Write an assembly code to input two unsigned and output the division
4. Write an assembly code to input two signed number and output the division
5. Take Decimal number as input and output the decimal number.

6. Your friend Swapnil collected different types of bank notes and coins. But now he is being very lazy to count the total amount of money he collected. So, you must help him count the total amount of money. But as you are an Computer Science Engineering student you want to do the job efficiently using an assembly program. Ignore the floating points.

Sample Input:

1000 taka note : 3
500 taka note : 5
200 taka note : 3
100 taka note : 7
50 taka note : 9
20 taka note : 2
10 taka note : 4
2 taka note : 3
50 paisa coin : 14
25 paisa coin : 13

Sample Output:

Total Money: 7346

7. Write a program in assembly to display a positive/negative decimal number (put in AX) as output. [using stack/not using stack]

8. Take an integer number as input and print the reverse of that number.

9. Write an assembly program that finds the sum of the following series up to n terms:
0, 1, 8, 27, 64, ... …

You must take n as an input, assuming 0<n<10. You need to put the final sum in the AX Register. Example: Input: n=5; Output: AX=225 (in decimal)



Thank You

QUESTION?



Stack & Procedure

Objectives

- Stack Operation
- Terminology of Procedures
- CALL and RET

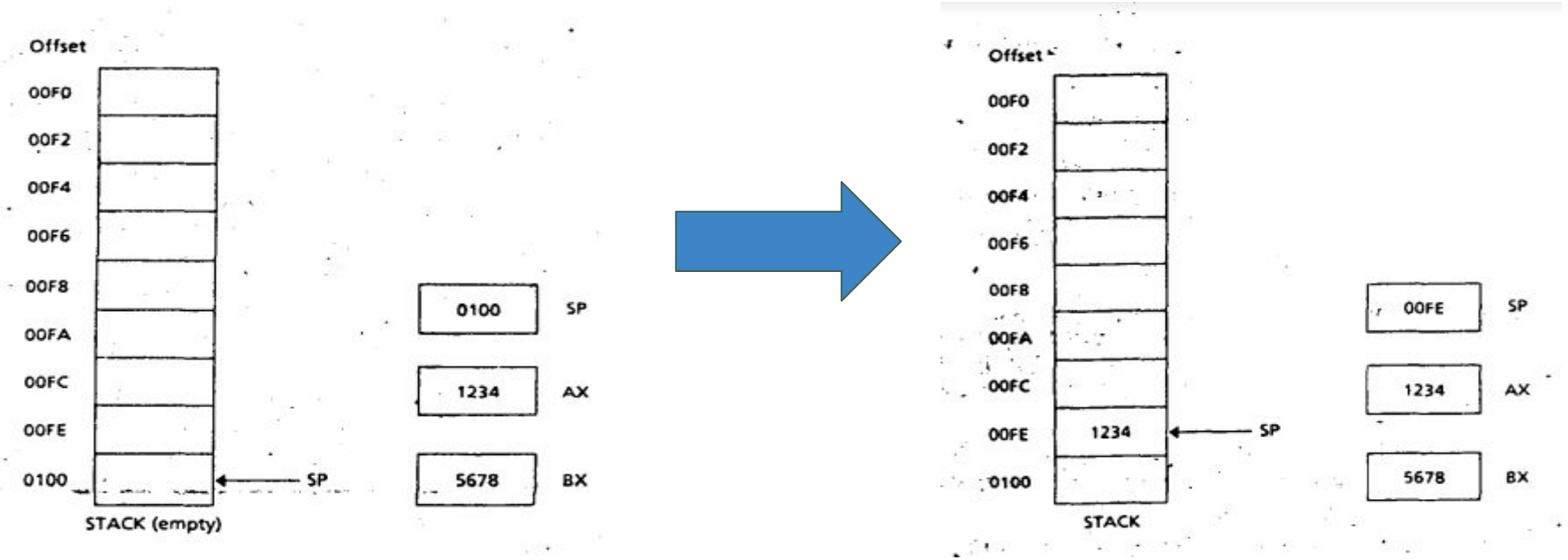
Book

Stack Operation: PUSH

- **Syntax:** `PUSH SOURCE` ; **16-bit** register or memory word.
- SP is decreased by **2**.
- A copy of the source content is moved to the address specified by SS:SP. The source is unchanged.

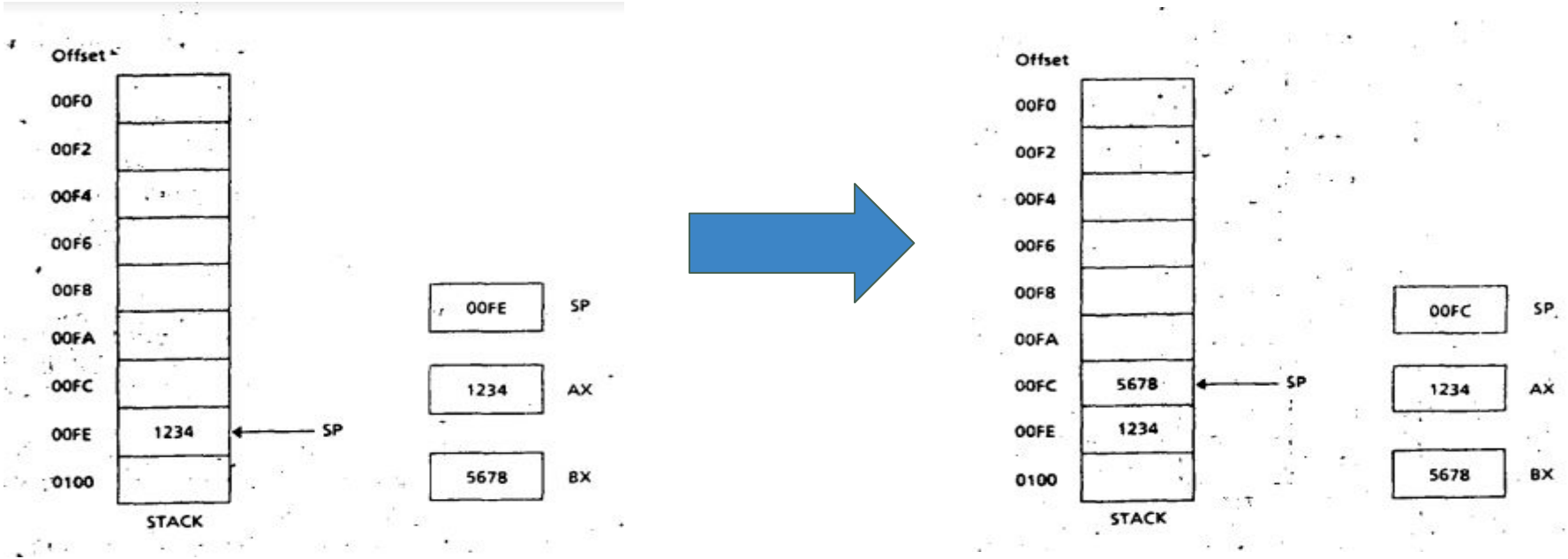
Stack Operation: PUSH

Instruction: **`PUSH AX`**



Stack Operation: PUSH

Instruction: **`PUSH BX`**

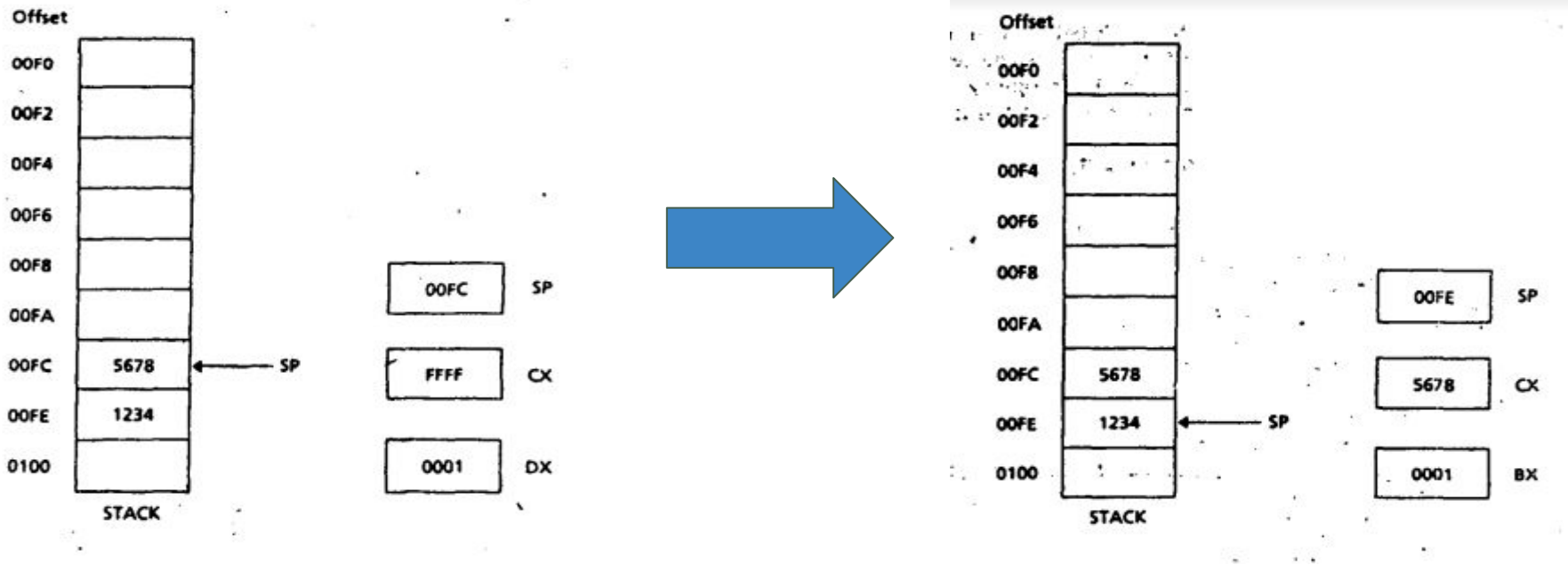


Stack Operation: POP

- **Syntax**: `POP DESTINATION` ; **16**-bit register (except IP) or memory word.
- The content of SS:SP (the top of the stack) is moved to the destination.
- SP is increased by **2**.

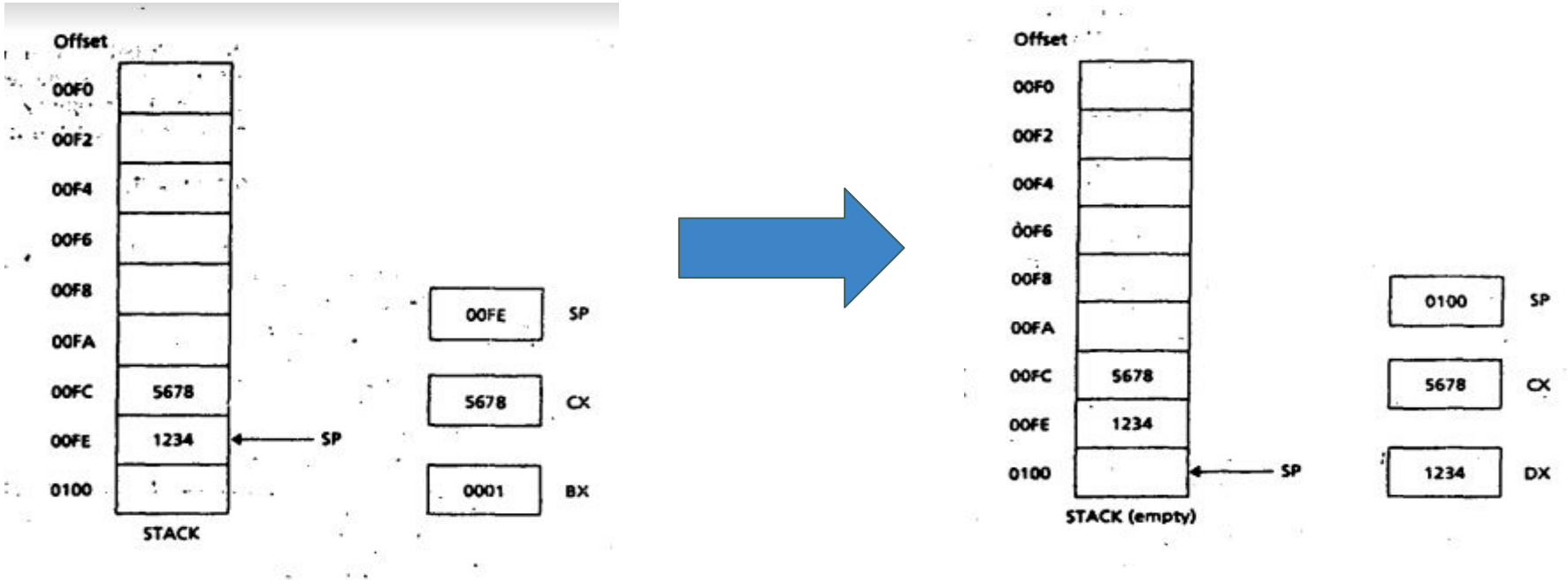
Stack Operation: POP

Instruction: `POP CX`



Stack Operation: POP

Instruction: **`POP DX`**



Stack Operation: Application

Algorithm to Reverse Input.

```
Display a '?'
Initialize count to 0
Read a character
WHILE character is not a carriage return DO
    push character onto the stack
    increment count
    read a character
END WHILE;
Go to a new 1ine
FOR count times DO
    pop a character from the stack;
    display it;
END FOR
```

Stack Operation: More

- **PUSHF**
- **POPF**
- Applications??

Procedure Declaration

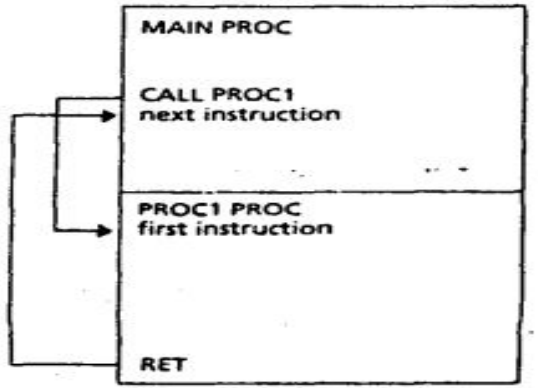
- **name** is the user-defined name of the procedure.
- The optional operand ***type*** is *NEAR* or *FAR* (if type is omitted, NEAR is assumed).
- *NEAR* means that the statement that calls the procedure is in the same segment as the procedure itself.
- *FAR* means that the calling statement is in a different segment.
- Assembly language procedures does not have parameter lists, so it's up to the programmer to devise a way for procedures to communicate. For example, if there are only a few input and output values, they can be placed in registers.

```
name PROC type
;body of the procedure
     RET
name ENDP
```

CALL and RET

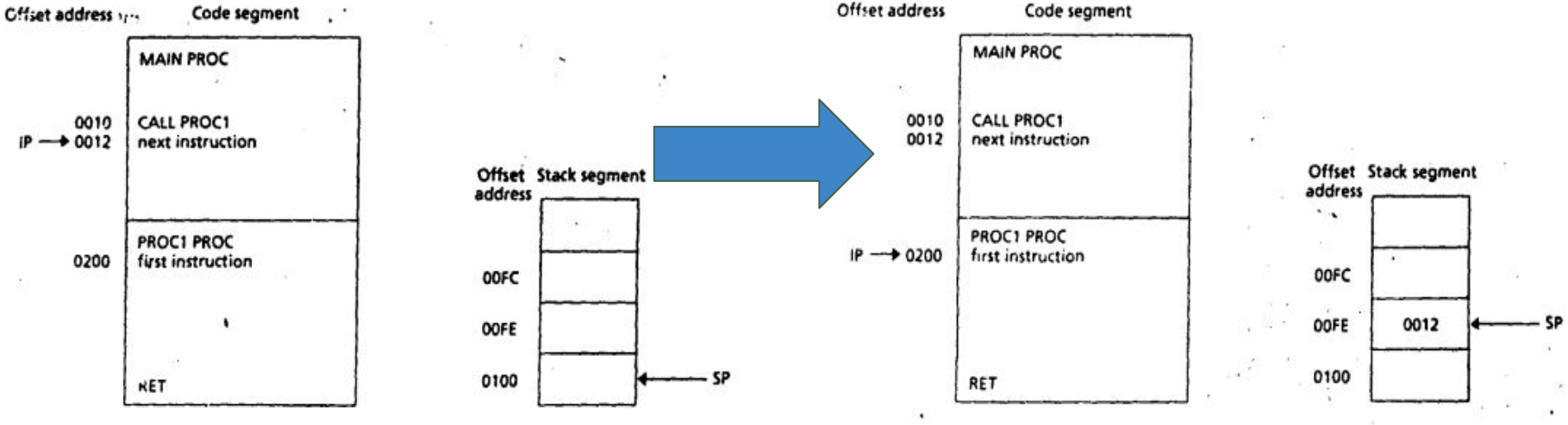
- To invoke a procedure, the CALL instruction is used.
  - `CALL name`
  - `CALL address_exprcssion`

- The `RET` (return) instruction causes control to transfer back to the calling procedure. Every procedure (except the main procedure) should have a RET someplace; usually it's the last statement in the procedure.



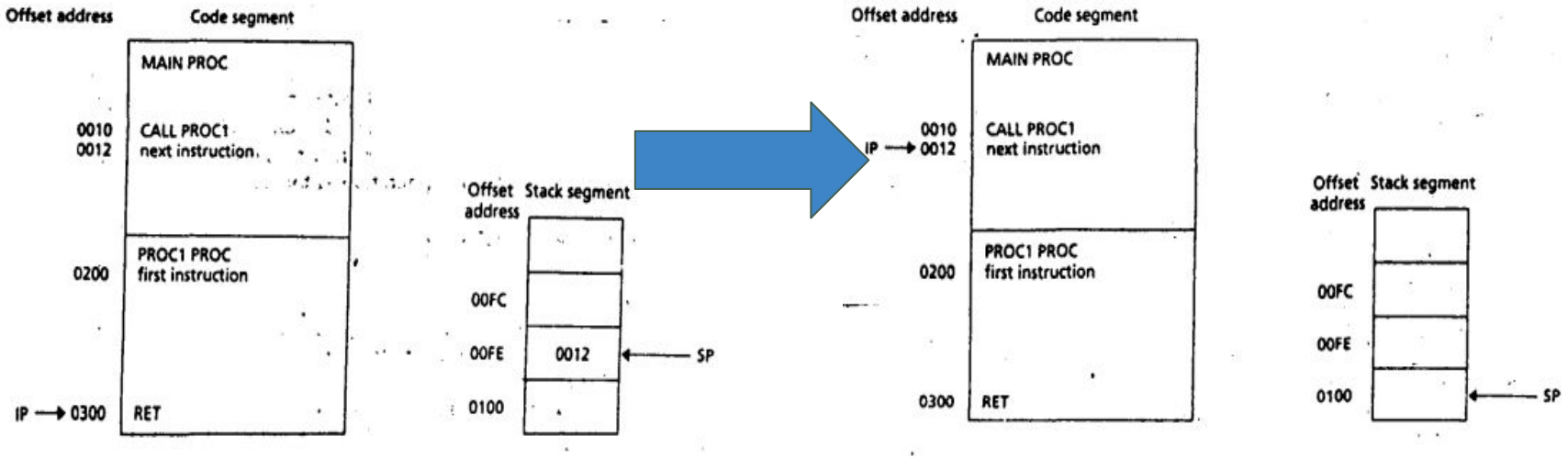
CALL and RET

- **`CALL name`**
  - Store the return address in stack.
  - IP changes to the address of the called procedure.



CALL and RET

- **RET**
  - POP out from the stack to IP·




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Thank You

Lec - 9
Arrays

Objectives

- Arrays
- Addressing Modes

Book

Arrays

- **Array Definition:** W DW 10, 20, 30, 40, 50, 60
- The address of the array variable is called the **base address** of the array. If the offset address assigned to W is 0200h, the array looks like this in memory:

Offset address	*Symbolic address*	*Decimal content*
0200h	W	10
0202h	W+2h	20
0204h	W+4h	30
0206h	W+6h	40
0208h	W+8h	50
020Ah	W+Ah	60

Arrays: The DUP Operator

- It is possible to define arrays whose elements share a common initial value by using the DUP (duplicate) operator.
- **Syntax:** repeat_count DUP (value)
- **Examples:**
- GAMMA DW 100 DUP (0) ;sets up an array of 100 words, with each entry Initialized to 0.
- DELTA DB 212 DUP (?) ;creates an array of 212 uninitialized bytes.

Array Declaration: C vs Assembly

```
ARRAY DB 5 DUP(?)      ; like: char arr[5];  (uninitialized)
NUMS  DB 10, 20, 30    ; like: int nums[3] = {10,20,30};  (initialized)
```

Array Element Access: C vs Assembly

```
LEA SI, ARRAY       ; SI = &arr[0]     (base address)
MOV AL, [SI]        ; AL = arr[0]      (dereference)
INC SI              ; SI = &arr[1]     (pointer++)
MOV AL, [SI]        ; AL = arr[1]
```

DW Array Element Access

```
LEA SI, NUMS
MOV AX, [SI]          ; nums[0]
ADD SI, 2             ; move by 2 bytes (word size), not 1
MOV AX, [SI]          ; nums[1]
```

Array Traversal: C vs Assembly

```
MOV CX, 5                    ; the "n"  loop counter
LEA SI, ARRAY                ; the "i=0"  start pointer
TRAVERSE:
   MOV AL, [SI]              ; arr[i]
   ; do required operation with array element (now in AL)
   INC SI                    ; i++ (pointer walks forward)
   LOOP TRAVERSE:            ; CX--, jump back if CX≠0
```

Assembly

```
for (int i=0 ; i<n ; i++)
{
        arr[i]
}
```

C

Addressing Modes

The way an operand is specified is known as its **addressing mode**. The addressing modes we have used so far are:

- **register mode**, which means that an operand is a register;
- **immediate mode**, when an operand is a constant; and
- **direct mode**, when an operand is a variable.

Examples:

MOV AX, 0 ;dest is in register mode but src is in immediate mode

ADD ALPHA, AX ;dest is in direct mode but src is in reg mode.

Addressing Modes: 8086

There are four additional addressing modes for the 8086:

(1) **Register Indirect**, (2) **Based**, (3) **Indexed**, and (4) **Based Indexed**. These modes are used to address memory operands indirectly. The first three of these modes are useful in one-dimensional array Processing.

Addressing Modes: Register Indirect

The **offset** address of the operand is contained in a register. The register acts as a pointer to the memory location.

Syntax: [register]

Registers: BX, SI, DI or BP.

Note: BP's segment number is in SS and others in DS.

Example: Suppose that **SI** contains 0100h, and the word at **0100h** contains 1234h.

MOV AX, [SI]

the CPU 1) examines SI and obtains the offset address 100h,

2) uses the address **DS:0100h** to obtain the value 1234h, and

3) moves 1234h to AX.

This is not the same as,

MOV AX,SI ;which simply moves the value of **SI**, namely 100h, into AX.

Addressing Modes: Register Indirect

Example 10.2 Suppose that

BX contains 1000h
SI contains 2000h
DI contains 3000h

Offset 1000h contains 1BACh
Offset 2000h contains 20FEh
Offset 3000h contains 031Dh

where the above offsets are in the data segment addressed by DS.

Tell which of the following instructions are legal. If legal, give the source offset address and the result or number moved.

```
a.  MOV  BX, [BX]
b.  MOV  CX, [SI]
c.  MOV  BX, [AX]
d.  ADD  [SI], [DI]
e.  INC  [DI]
```

Solution:

	Source offset	*Result*
a.	1000h	1BACh
b.	2000h	20FEh
c.	illegal source register	(must be BX, SI, or DI)
d.	illegal memory–memory addition	
e.	3000h	031Eh

Now let's return to the problem of adding the elements of an array.

Addressing Modes: Based and Indexed

The operand's offset address is obtained by adding a number called a displacement to the contents of a register. Displacement may be any of the following:

1. the offset address or a variable
2. a constant (positive or negative)
3. the offset address or a variable plus or minus a constant.

Syntax:

[register + displacement]

[displacement + register]

[register] + displacement

displacement + [register]

displacement [register]

Registers: BX, SI, DI or BP.

Note: BP’s segment number is in SS and others in DS.

The addressing mode is called **based** if BX (base register) or BP (base pointer) is used; it Is called **indexed** if SI (source index) or DI (destination index) is used.

Addressing Modes: Based and Indexed

Example 10.6 Suppose that ALPHA is declared as

```
ALPHA   DW   0123h,0456h,0789h,0ABCDh
```

in the segment addressed by DS. Suppose also that

BX contains 2
SI contains 4
DI contains 1

Offset 0002 contains 1084h
Offset 0004 contains 2BACh

Tell which of the following instructions are legal. If legal, give the source offset address and the number moved.

a. `MOV AX,[ALPHA+BX]`
b. `MOV BX,[BX+2]`
c. `MOV CX,ALPHA[SI]`
d. `MOV AX,-2[SI]`
e. `MOV BX,[ALPHA+3+DI]`
f. `MOV AX,[BX]2`
g. `ADD BX,[ALPHA+AX]`

Solution:

	Source offset	*Number moved*
a.	ALPHA+2	0456h
b.	2+2 = 4	2BACh
c.	ALPHA+4	0789h
d.	−2+4 = 2	1084h
e.	ALPHA+3+1 = ALPHA+4	0789h
f.	Illegal form of source operand	
g.	Illegal source register	

The next two examples illustrate array processing by based and indexed modes.

Tasks

1. Find the sum of all the elements of the array arr that contains 10, 20, 30, 40, 50
2. Take an integer n as the array size and input n elements into the array. Finally, print the sum of all the elements in the array. (Solution)
3. Given an array that contains 10, 11, 9, 15, 20, 25, 21, find the maximum element from the array. (solution)
4. Separate the even and odd elements of an array and store them in two arrays named even and odd, respectively. Print both even and odd arrays. (Solution)
5. Insert an element to an array at a given position.
6. Merge two sorted arrays into one sorted array
7. Given an array that contains 5, 4, 6, 2, 3, 7, 1, 7, reverse the array without using any additional array (two pointer approach). (Solution)
8. Calculate frequency of each element for the given an array of 5, 6, 5, 6, 5, 7, 8 values.
9. Sort an array using bubble sort.

Procedure

- Procedure is a reusable, named block of code you can jump into, execute, and return from
- Main Syntax

```
name PROC
    ; body
    RET
name ENDP
```

Procedure Definition

```
Call name
```

Procedure Calling

Procedure: CALL and RETURN

- **CALL name**
  - PUSH the return address in stack.
  - IP changes to the address of the called procedure.

- **RET**
  - POP out from the stack to IP·

Procedure: Parameters

- Arguments are not “sent” to procedures in Assembly. They are either
  - stored in registers (AX, BX etc)
  - or pushed onto stack
- The procedure is written accordingly to take the required parameters from the registers or the stack

```
SUM PROC
    ADD AX , BX
    RET
SUM ENDP
```

```
MOV AX, 5
MOV BX, 6
CALL SUM
```

Procedure: Return Values

- Similarly return values are not “sent back” to calling procedures in Assembly. They are
  - stored in registers (AX, BX etc)
- The calling procedure (usually main proc) is written to access the return values from the registers used

```
SUM PROC
    ADD AX + BX
    RET
SUM ENDP
```

```
MOV AX, 5
MOV BX, 6
CALL SUM
; Required sum is in AX
```

Tasks

1. **Character Case Converter:** Change a character's (stored in AL) case and store it in AL using a procedure. Take the character input from main procedure. Print the character before and after case conversion in main procedure. (Solution)
2. **Maximum Element:** Find the maximum element from an array and store it in AX using a procedure. Take the array input in one procedure, and the maximum finding logic in another procedure. Call the procedures from main procedure
3. **Count Even Elements:** Write a procedure CountEven that takes an array's offset (SI) and length (CX), and returns the count of even numbers in the array in AX.
4. **Array Reversal:** Write a procedure ReverseArray that takes an array's offset (SI) and length (CX), and reverses the array in place. (Solution)
5. **Linear Search:** Write a procedure LinearSearch that takes an array and a target value (AX), and returns the index of the target in BX if found, or -1 in BX if not found.


THANK
YOU

Micro-controllers

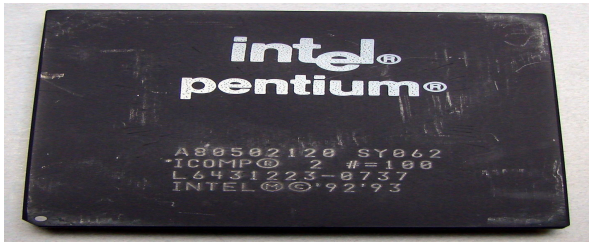
Topics

- Microprocessors, Micro-controllers
- Arduino
- Pin Diagram

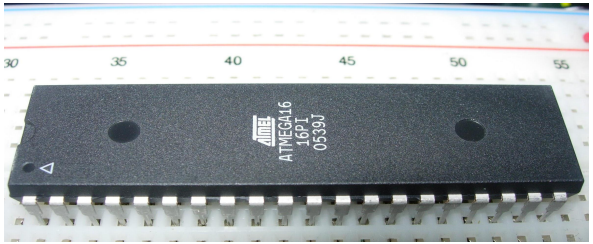
Book: https://drive.google.com/file/d/1Z_1xdJuqq9HiUi07VAzyjcTnhyGeqib7/view

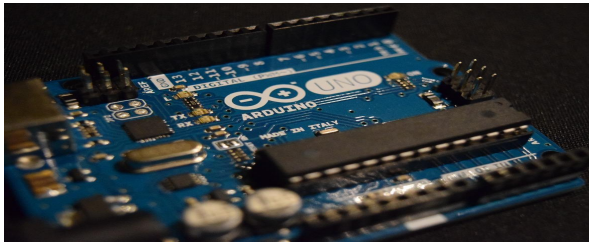
Microprocessors, Micro-controllers





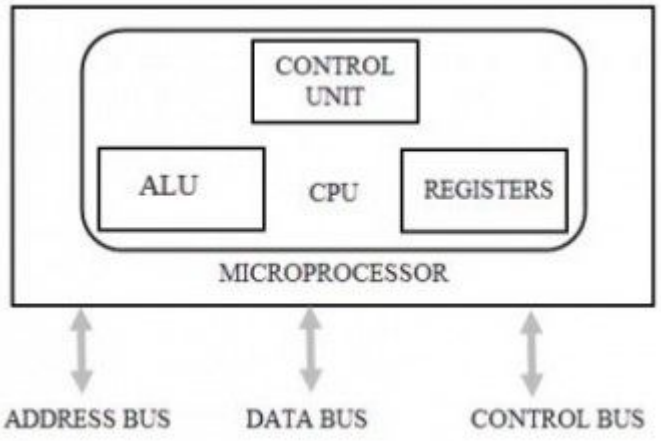
Microprocessors



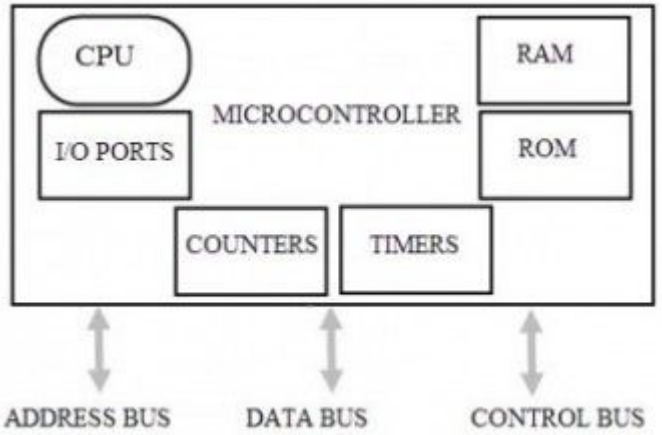


Micro-controllers

Microprocessors, Micro-controllers



Microprocessors



Micro-controllers

Microprocessors, Micro-controllers

- Microprocessor consists of only a CPU and Microcontroller contains a CPU, Memory, I/O all integrated into one chip.
- Microprocessor is used in PCs and Microcontroller is used in an embedded system.
- Microprocessor is complicated and expensive but Microcontroller is inexpensive and straightforward.

Proteus

- Proteus is a electrical suite for circuit simulation purposes.
- Proteus Download Link: https://drive.google.com/file/d/1aQ-QefxwPn7Coc0X_xKwAKFt6r3PQLiD/view
- Wokwi: https://wokwi.com/
- Arduino IDE: https://downloads.arduino.cc/arduino-nightly-windows.zip
- Arduino Uno datasheet: https://www.farnell.com/datasheets/1682209.pdf
- Arduino Uno figure: https://drive.google.com/file/d/1Vt29W6dWGlfW3rGKwDYeUG2LrjclE05N/view?usp=sharing
- Functions: https://www.arduino.cc/reference/en/

Arduino Uno

- Arduino is an open-source hardware and software company that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices.
- The Arduino Uno is a microcontroller board based on the ATmega328P.
- A ceramic resonator generates the ATmega328P's clock signal, though this oscillator features less precision than its crystal-based cousin.

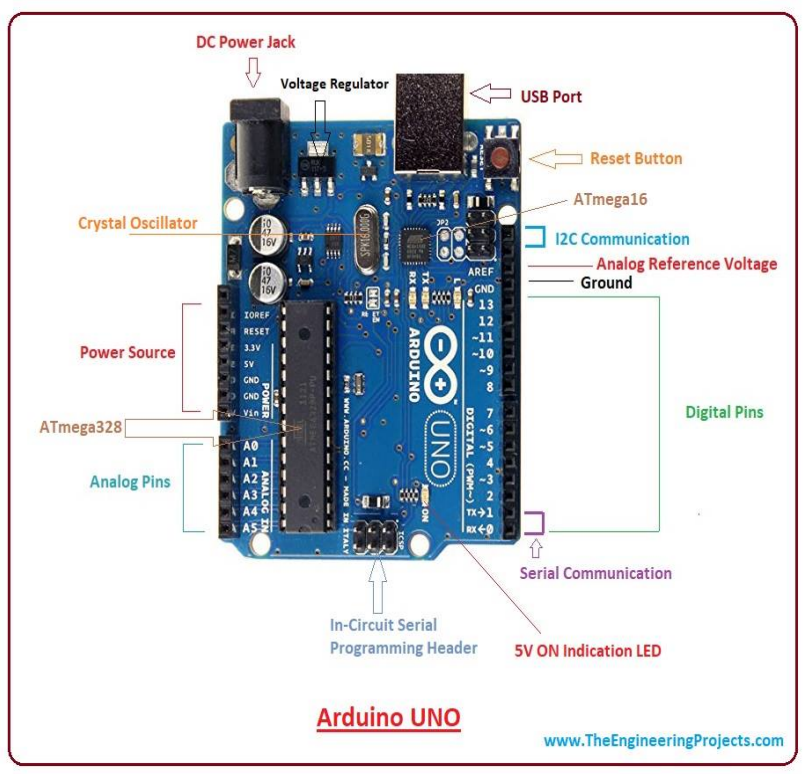
Few Sensors

- Analog Sensors
  - Resistance-based sensors
    - LDR (Light-dependent resistor)
    - RTD (Resistance-based Temperature Detectors)
      - P/NTC (Positive/ Negative Temperature Calibration)
    - Position Sensors → Joystick
  - Voltage-based sensors
    - LM35 temperature sensor (The voltage difference is proportional to the temperature.)
  - Current-based sensors
    - Phototransistors/ Photodiode

Arduino Uno

Pin Diagram

- 14 digital input/output pins (of which 6 can be used as PWM outputs)
- 6 analog inputs (also can be used as digital I/O)
- One UART interface found on pin 0 (RX0) and pin 1 (TX0)
- One I2C/TWI module
- And a SPI bus



Arduino Uno Input-Output

- Digital I/O
  - **Pins**: 0 to 13 + Analog Pins
  - **Req**: pinMode(pin, mode)
    - mode: INPUT/ OUTPUT
  - **Input**: digitalRead(pin)
  - **Output**: digitalWrite(pin, value)
  - Examples:
  - https://wokwi.com/projects/335733385937814098
  - https://wokwi.com/projects/335736198446187091

Arduino Uno Input-Output

- Analog Input
  - **Pins**: A0 to A5
  - **Technique**: Use 10 bits ADC
  - **Input**: analogRead(pin)
    - Return value: 0-1023 (10 bits)
  - **analogReference(type);**
    - DEFAULT: On Board (5 or 3.3V)
    - EXTERNAL: Applied on AREF pin (0 to 5V only)
  - Discrete value to Analog Value
    - analogVoltage = (digitalValue * V_{ref}) / 1023.00;
    - analogVoltage = map(digitalValue, fromLow, fromHigh, toLow, toHigh);
      - fromLow = 0, fromHigh = 1023
      - toLow = 0, toHigh = V_{ref}
  - Example: https://wokwi.com/projects/335731558488998483

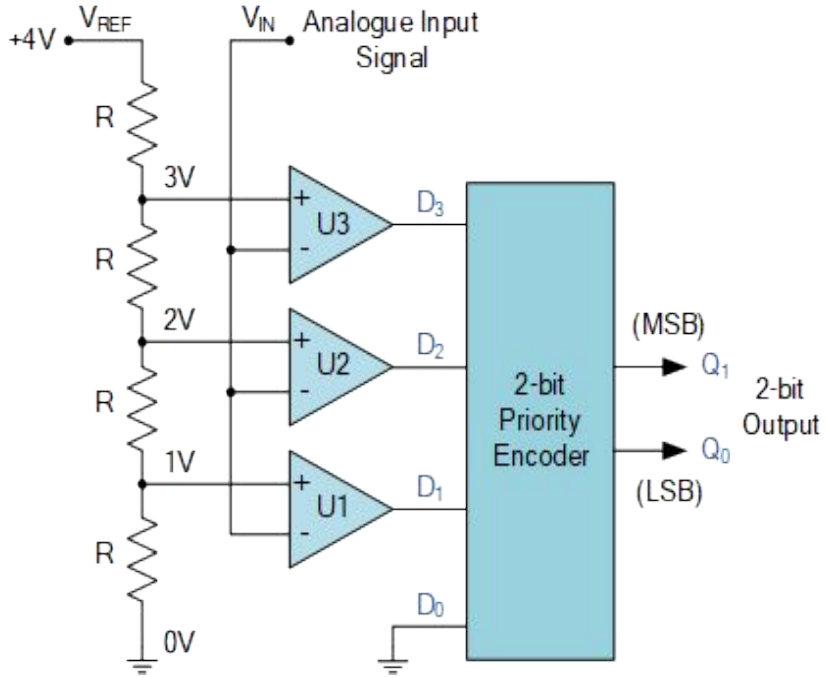
FLASH ADC

Analog Inputs: 2-bit Analog to Digital Converter Circuit

- Analog signals can be continuous and provide an infinite number different voltage values.
- Digital circuits on the other hand work with binary signal which have only two discrete states, a logic “1” (HIGH) or a logic “0” (LOW).
- Analog signals come from various sources and sensors which can measure sound, light, temperature or movement.
- Many actuators may need analog signals for controlling purposes like motor speed.

Analog Inputs: 2-bit Analogue to Digital Converter Circuit

Comparator Outputs				Digital Outputs	
D_3	D_2	D_1	D_0	Q_1	Q_0
0	0	0	**0**	0	0
0	0	**1**	X	0	1
0	**1**	X	X	1	0
1	X	X	X	1	1



2-bit ADC will give digital value from 0 to 2^2 - 1 = 3, total 4 values. Hence for 10-bit it is 1024(0 to 1023).

Analog Inputs: Designing n-bit Flash ADC

- Requirements for n-bit Flash ADC:
  - Number of Comparators: $2^n - 1$
  - Number of Resistance: 2^n
  - 2^n to n Priority Encoder
  - V_{ref} and V_{in}
  - Compare V_{in} and V_{xR}
    - V_{in} is Analog Input Signal
    - V_{xR} is the voltage after xth resistance

Analog Inputs: Digital Output value Calculation

- **ADC Resolution (1 LSB) = V_{ref} / (2^n)**
  - The smallest incremental voltage that can be recognized.
- **Digital value = (V_{in} / V_{ref}) * (2^n – 1)**
- **Aout = (Digital value * V_{ref}) / (2^n – 1)**

Where,

Vref - The reference voltage is the maximum value that the ADC can convert.[Uno has a **Vref pin**]

To keep things simple, let us consider that Vref is 5V,

For 0 Vin, digital o/p value = 0

For 5 Vin, digital o/p value = 1023 (10-bit)

Analog Inputs: Problems

Design a 3-bit flash ADC using the followings:

- Resistors
- Comparators
- Priority Encoder

For both 3-bit and 10-bit ADC calculates the followings (Suppose Vref is 5v):

- ADC Resolution (1 LSB)
- Digital Value when Vin = 3.3
- Aout from the digital value

Analog Inputs: Temperature Sensor (LM35)

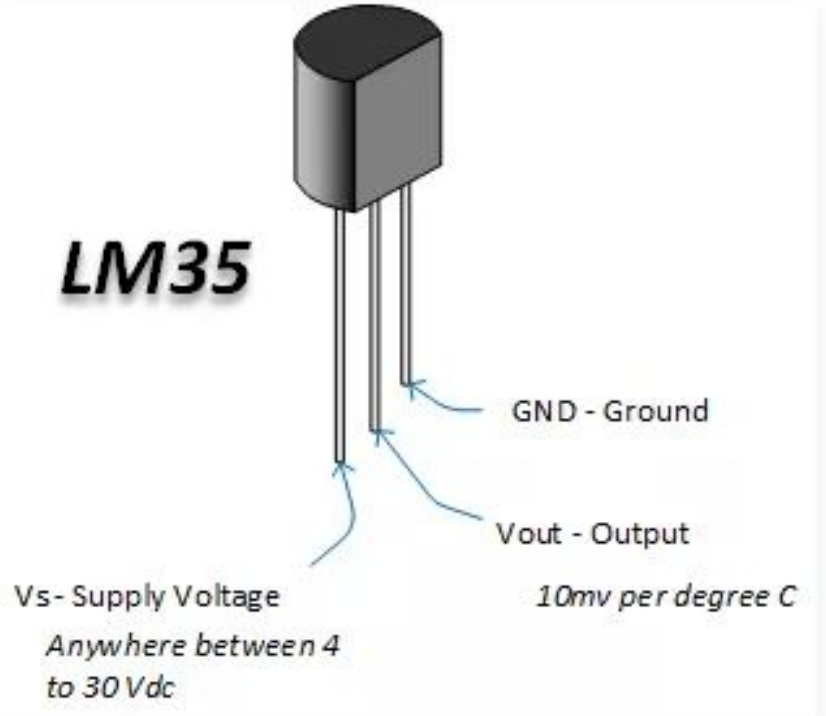
It’s a temperature device with an output voltage linearly proportional to the Centigrade temperature. Temperature range: −55°C to 150° C. The sensitivity of LM35 is 10 mV/degree Celsius.

Pin Connection:

- First pin(Vs) on the left to 4-30V power,
- Second pin(Vout) to any analog input pins,
- Third pin(GND) to the ground

Formula,

- temp = (1/ 2.048) * analogRead(Vout)

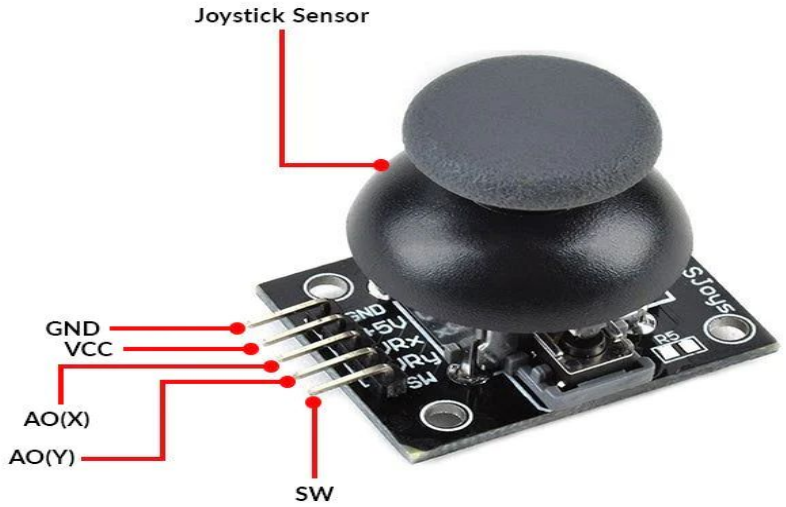


Analog Inputs: Analog Joystick

- Joystick is an input device. It is used to control the pointer movement in 2-dimension axis.
- The joystick has two potentiometers to read user's input.
- One potentiometer is used to get the analog output voltage for X-Direction movement and the other potentiometer is used to get the analog output voltage for Y-Direction movement.
- At Idle position output voltage, will be VCC/2.

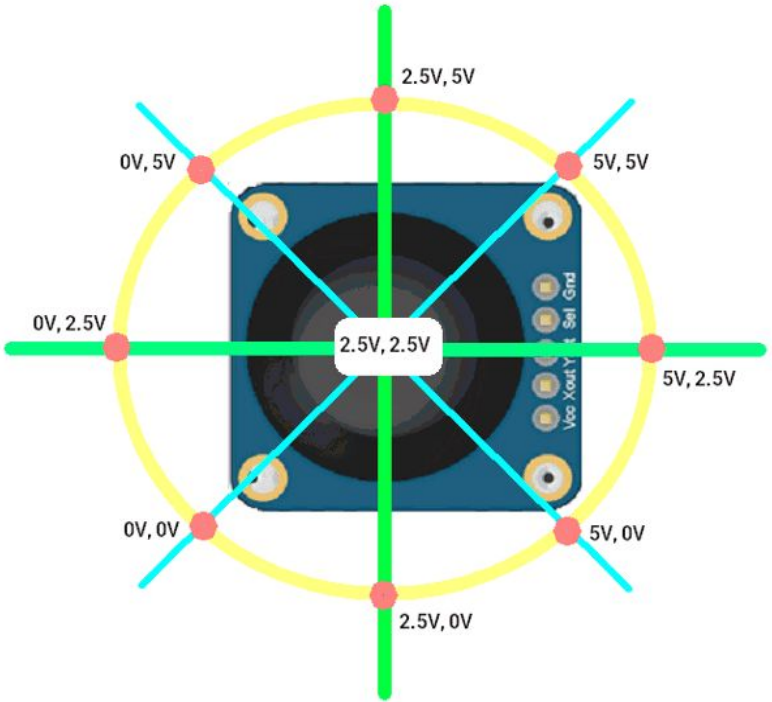
Pin Connection:

- VCC: First pin(Vs) on the left to 4-30V power,
- GND: Ground
- AO(X): A0 to A5
- AO(Y): A0 to A5



Analog Inputs: Analog Joystick

- Formula:
  - Left:
  - Right:
  - Up:
  - Down:
- Example: https://wokwi.com/projects/335776852784185938



Arduino Uno Input-Output

- Analog Output
  - Pins: 3, 5, 6, 9, 10, 11
  - Technique: PWM
  - Output: analogWrite(pin, value)
    - value: the duty cycle: between 0 (always off) and 255 (always on). Allowed data types: int.
    - PWM voltage=(Duty cycle ÷ 255) x 5 V.
  - Used to control the intensity of the transducer like led, motor.
  - Example: https://wokwi.com/projects/335737308028338770

Pulse Width Modulation

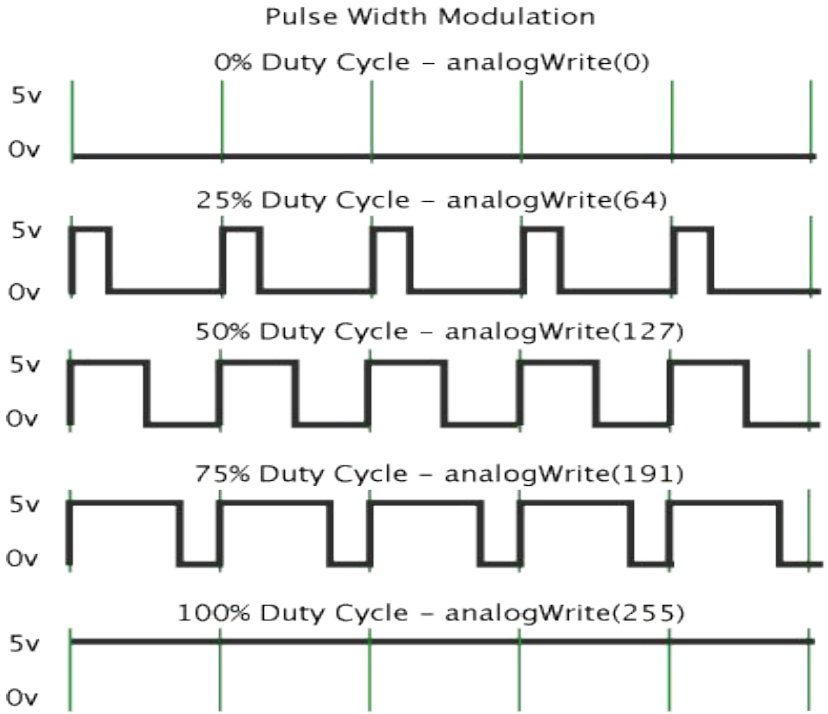
Analog Outputs: PWM (Pulse Width Modulation) ‘~’

Arduino does not have in-board DAC. Instead, it uses PWM(‘~’).

Can be used in:

- **Controlling the brightness of LED**
- Speed control of DC motor
- Controlling a servo motor or
- Where you have to get analog output with digital means

PWM voltage=(Duty cycle ÷ 255) x 5 V.



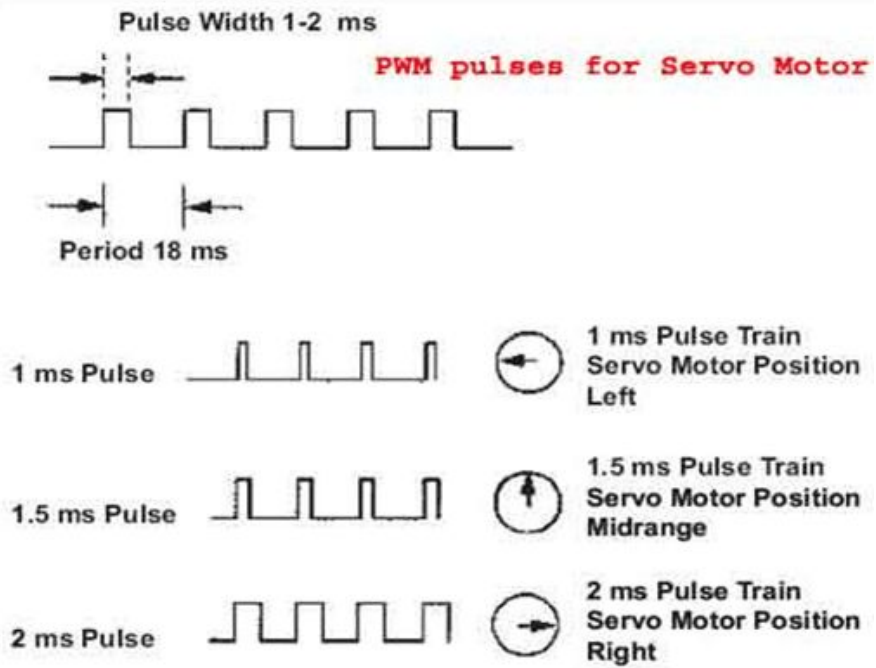
Analog Outputs: Servo Motor

- **Pins:**
  - VCC
  - GND
  - PWM



Analog Outputs: Servo Motor

- **Servo Motor:**
  - Servo checks the pulse in every 20 milliseconds.
  - The pulse of 1 ms (1 millisecond) width can rotate the servo to 0 degrees, 1.5ms can rotate to 90 degrees (neutral position) and 2 ms pulse can rotate it to 180 degree.



Examples: https://wokwi.com/projects/335916736176980563

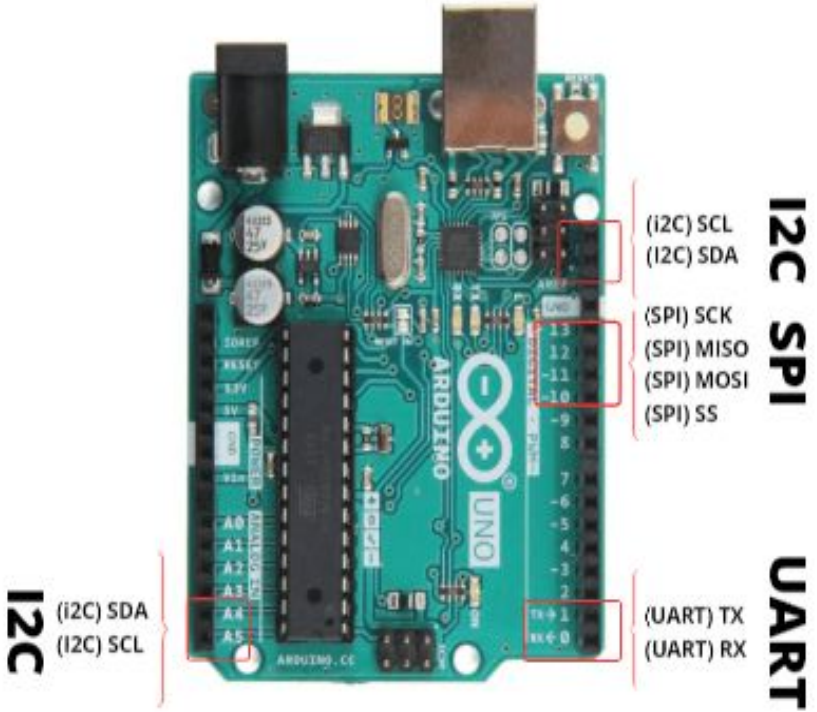
Serial Communication

Arduino Communication Peripherals: Serial Communication

- UART
- I2C
- SPI

Arduino Communication Peripherals: Serial Communication

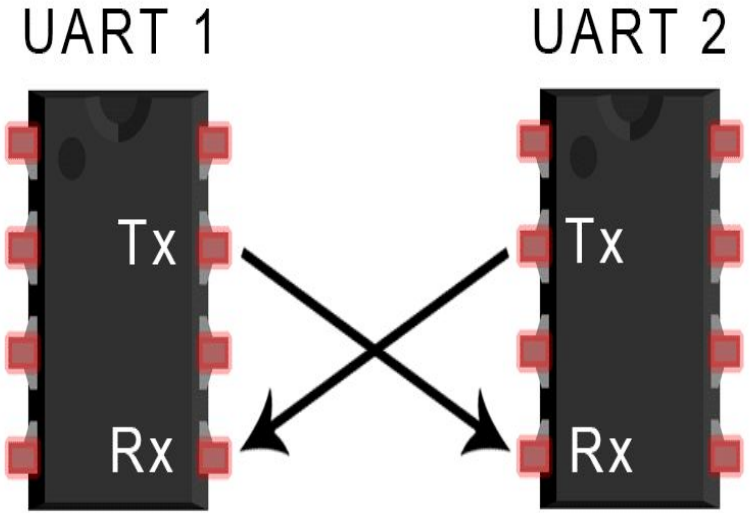
- UART
- I2C
- SPI



UART(Universal Asynchronous Reception & Transmission)

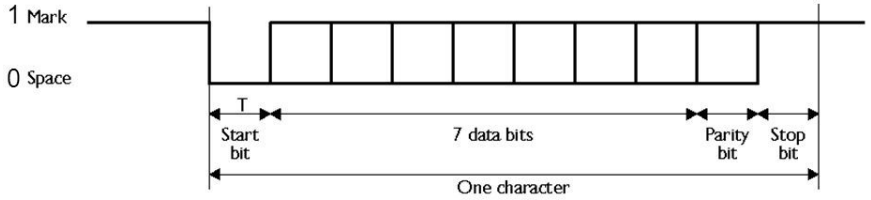
- Simple serial communication protocol that allows the host (Arduino) to communicate with serial devices [Pin 0 & Pin 1] by using Serial.print(“---”) instructions.
- Supports **Bidirectional**, **Asynchronous** and Serial Data Transmission
- **No clock** needed, uses **Synchronization Bits**, **Data Bits**, **Parity Bits**, **Baud Rate**
- **SoftwareSerial** library has been developed to allow serial communication on other digital pins of the Arduino

UART(Universal Asynchronous Reception & Transmission)

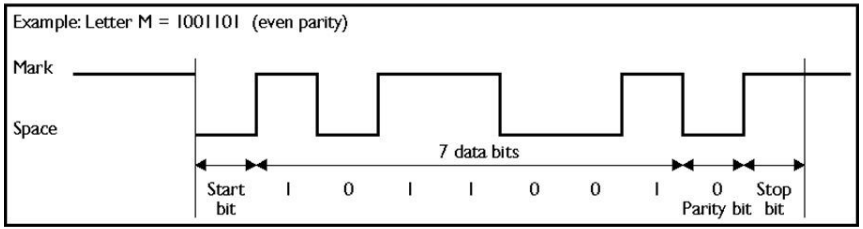


Connection

Asynchronous serial transmission



Serial transmission is little endian (least significant bit first)



7-4

Communication

Serial Communication (UART): Hardware Mechanism

- Two ways to communicate using UART :
  - Hardware Mechanism
  - Software Mechanism
- Hardware Mechanism uses 2 pins
  - RX - Pin 0 - Used for receiving
  - TX - Pin 1 - Used for transmitting
  - We can also use more than one UART using software mechanism
  - Computer via the USB port (Serial Monitor/Virtual Terminal)
    - Onboard USB-to-Serial converter (Atmega16U2)

Serial Communication (UART): Hardware Mechanism

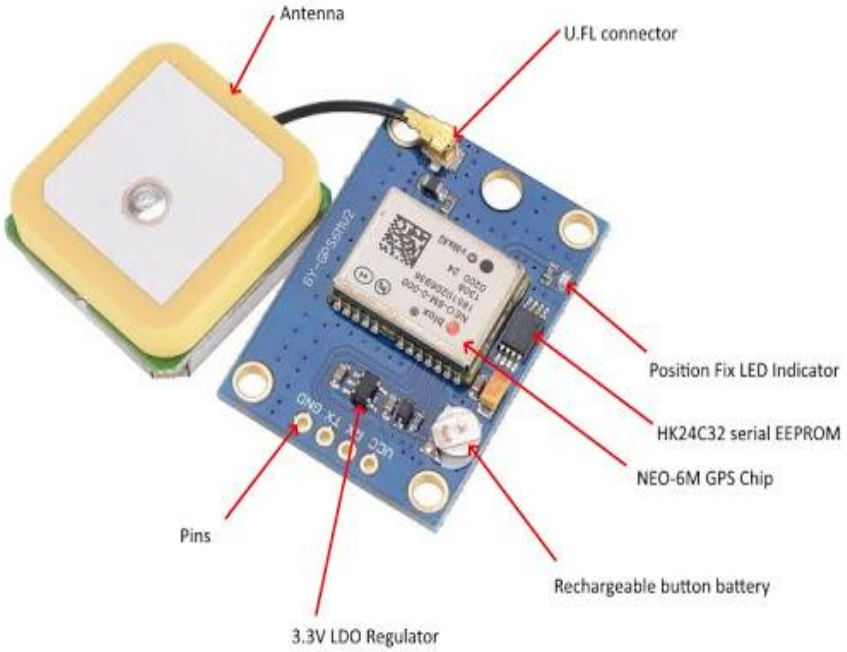
- By **Serial** command we can use UART pins.
- Some Basic UART Functions
  - Serial.begin(Baud rate)
  - Serial.end()
  - Serial.read()
  - Serial.write(parameters)
    - val //1 byte
    - string //series of bytes
    - buf, len//an array of len
- Example: https://www.tinkercad.com/things/0w8iOvi3soz-uart/editel

Serial Communication (UART): Software Mechanism

- Arduino IDE has a built-in software serial library which allows use to perform serial communication using other digital input-output pins.
  - #include <SoftwareSerial.h>
- Create an instant or object (often in global section).
  - SoftwareSerial <obj name> (RX, TX);
  - SoftwareSerial UART0(2, 3);
- To use the UART functions we have to use the instance.
- We can use the UART functions as the member function of the instance.
  - UART0.begin(9600);

Serial Communication (UART): GPS (Global Positioning System) Module

- Connect through serial communication
- Communicate with the Arduino using the UART pins
- Transmit information in NMEA format
- Need TinyGPS library to decode the information



NEO-6M GPS module

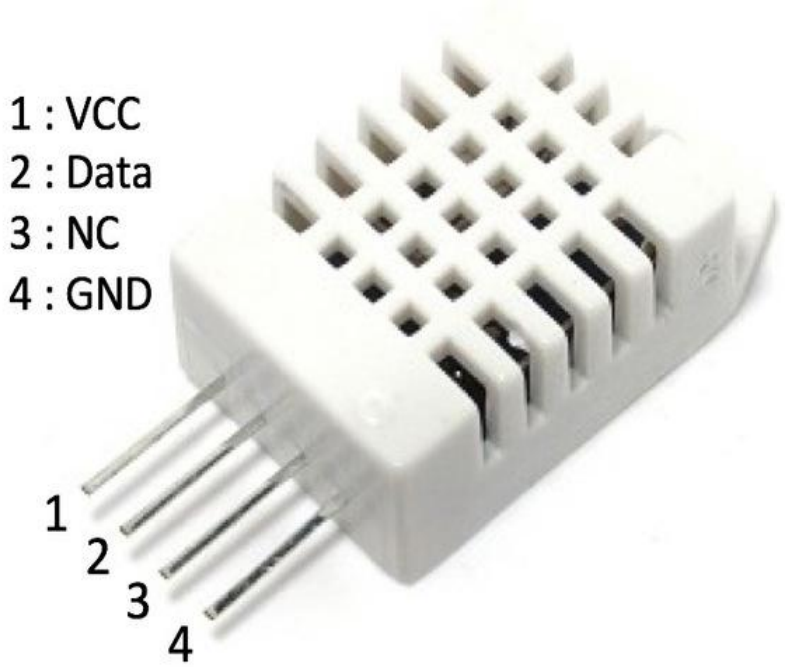
Humidity Sensor (DHT22)

It’s a digital-output, relative humidity, and temperature sensor.

Pin Connection:

- First pin(VCC) on the left to 3-5V power,
- Second pin(Data) to any digital input pins,
- Third pin(NC) is no-connect pin,
- The right-most pin(GND) to the ground

Link:
https://wokwi.com/projects/335781020271903315



I^2C(Inter-Integrated-Circuit)

- Bidirectional two-wire **Synchronous Half-duplex** serial bus and only 2 wires (SDA and SCL) to transmit information between devices connected to the bus
- Can communicate between up to almost 128 (112) slave devices using 7 bits addressing
- Supports multiple master devices and Two signals − SCL and SDA
  - SCL(A5) is the clock signal, and
  - SDA(A4) is the data signal
- Master can choose any slave device by its unique address
- Need to include “**Wire.h**” library
- Example: https://wokwi.com/projects/336327908183245396

SPI(Serial Peripheral Interface)

- Bidirectional, **Synchronous, Full-duplex** serial bus and 4 wires to transmit information between devices connected to the bus:
  - SCK[13] − This is the serial clock driven by the master.
  - MOSI[11] − This is the master output / slave input driven by the master.
  - MISO[12] − This is the master input / slave output driven by the master.
  - SS[10] − This is the slave-selection wire.
- Compared to UART and I2C, it is the fastest communication peripheral but doesn't support multiple master
- Master can choose any slave device by slave select pin
- Need to include “**SPI.h**” library

Mini Projects

Control a Servo with a Potentiometer

The code and the schematic diagram is given in the link:

https://wokwi.com/projects/442451897834479617

Automated Traffic Control System using Arduino for the Intersection of Four Roads

The code and the schematic diagram is given in the link:

Project Link: https://wokwi.com/projects/442453604887084033

Thank You

